

Lecture 15: Maps

Heather Mattie, PhD

October 21, 2025

Recipe of the Day!

Diwali Dinner Recipes

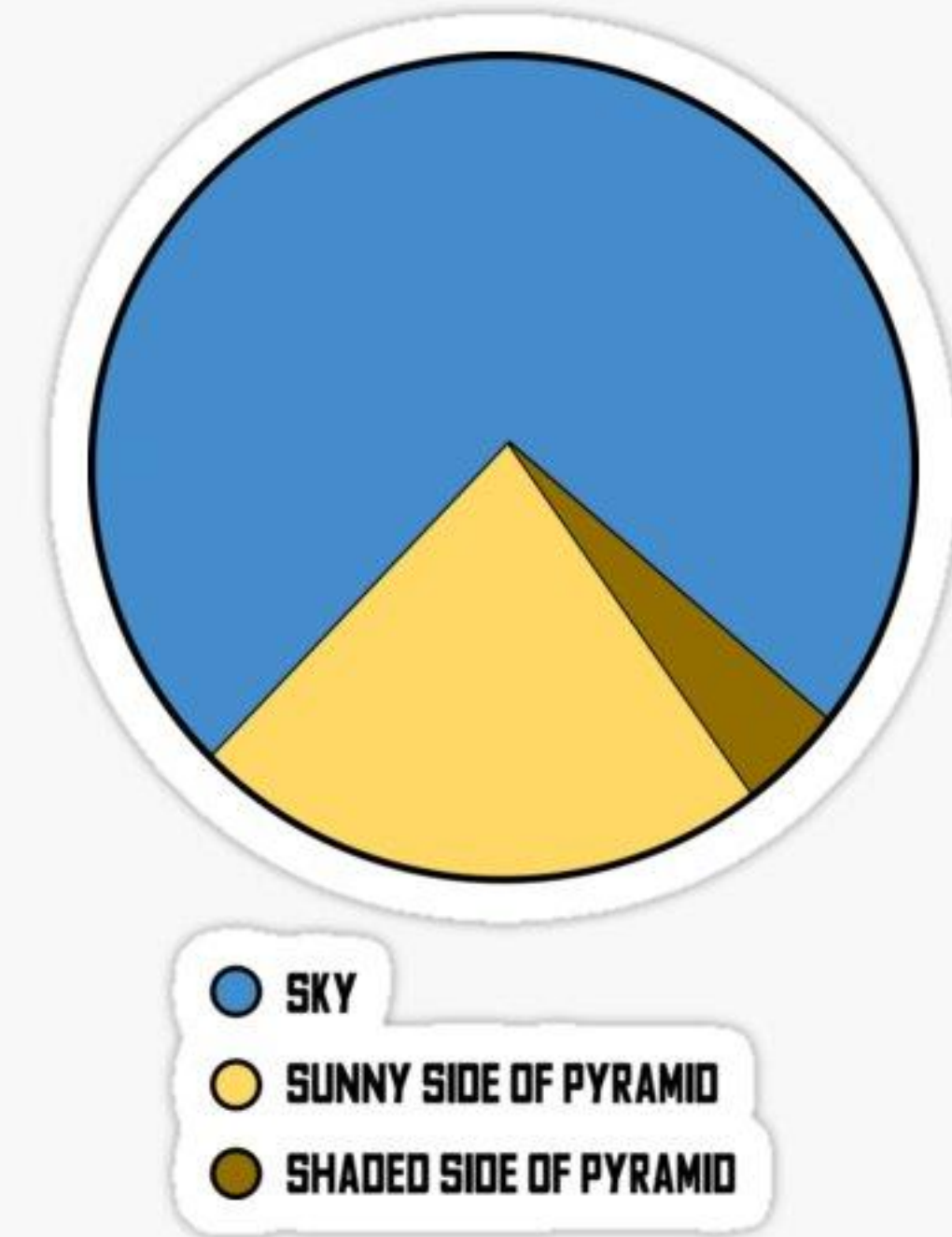


Connors Farm
Danvers, MA



Announcements

- Homework #3 due Friday, October 24 by 11:59pm
- No lab this week
- Midterm will be released on **Monday, October 27** and due **Friday, November 7**
 - No late days may be used
 - Generative AI **may not** be used in **any way**



Homework reminders

- If something is unclear, please ask in office hours, on canvas, or via email
- Generative AI may be used to help with debugging or explaining a concept, but you should not copy and paste code from it and need to cite when and how you used it

Mid-course survey results

- 45% response rate
- Useful for learning
 - In-class coding
 - Lectures and lab are recorded
 - Lecture slides, RMarkdown notes
 - Pacing
 - Learning environment

Mid-course survey results

- Areas for improvement
 - Too much time spent on debugging student code in class
 - Too much time spent setting up GitHub in class
 - More coding examples and/or cheat sheets
 - Mentioning which lectures should be referenced for which homework assignment
 - Rubric for homework assignments
 - More guidance on how to use Rmd files in course repository
 - Less time for slides, more time for in-class coding

Coding question of the day

Using the `gapminder` dataset, make a line graph of life expectancy over time. Include one line per country, and another line with the global average life expectancy.

Be sure to label the global average line and to make it stick out among the country-specific lines. Use the code below to calculate the crude global average for each year.

Hints:

1. Use `alpha` to help make the global average line stick out.
2. Use 2 `geom_line` layers.
3. Note that `avg` is a data frame.

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data("gapminder")

avg <- gapminder %>%
  group_by(year) %>%
  summarize(global_avg = mean(life_expectancy))
```

Maps

Maps Roadmap

1. The **maps** package
2. Plotting global life expectancy
3. Plotting US state murder rates
4. Case study: COVID-19 cases

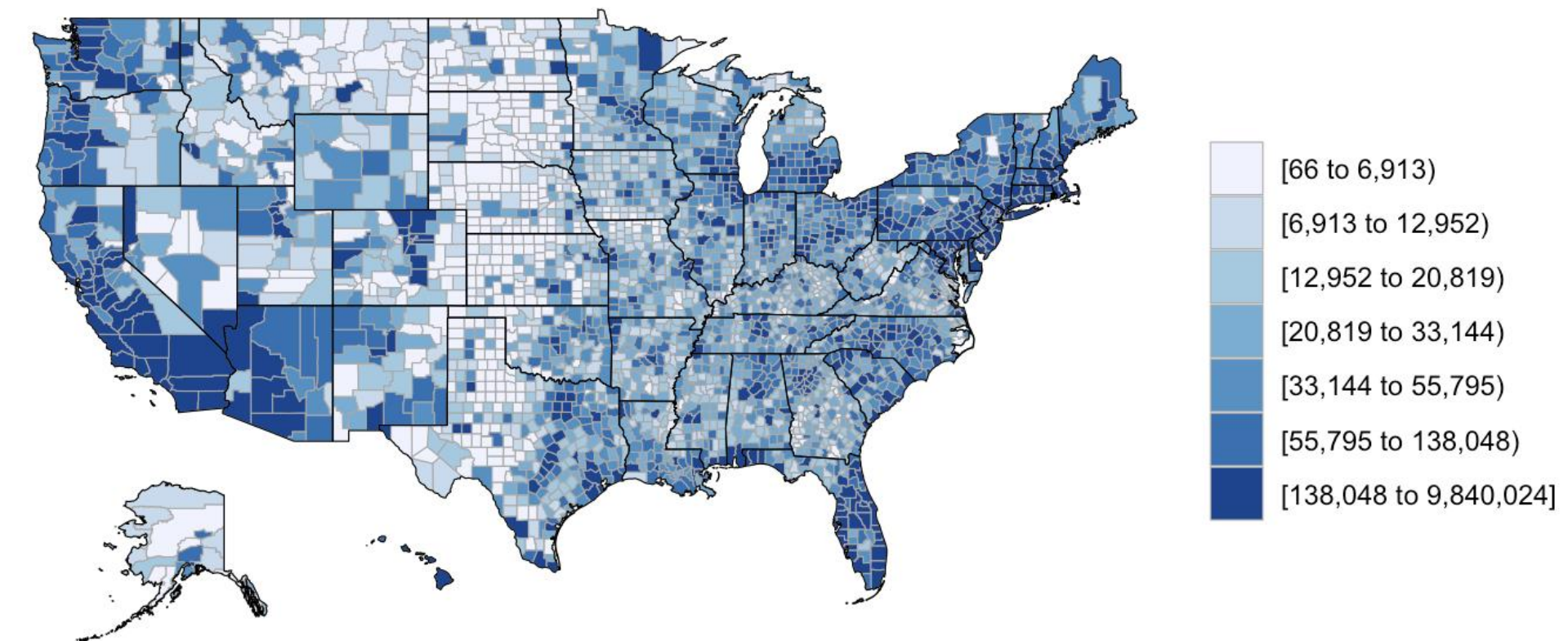
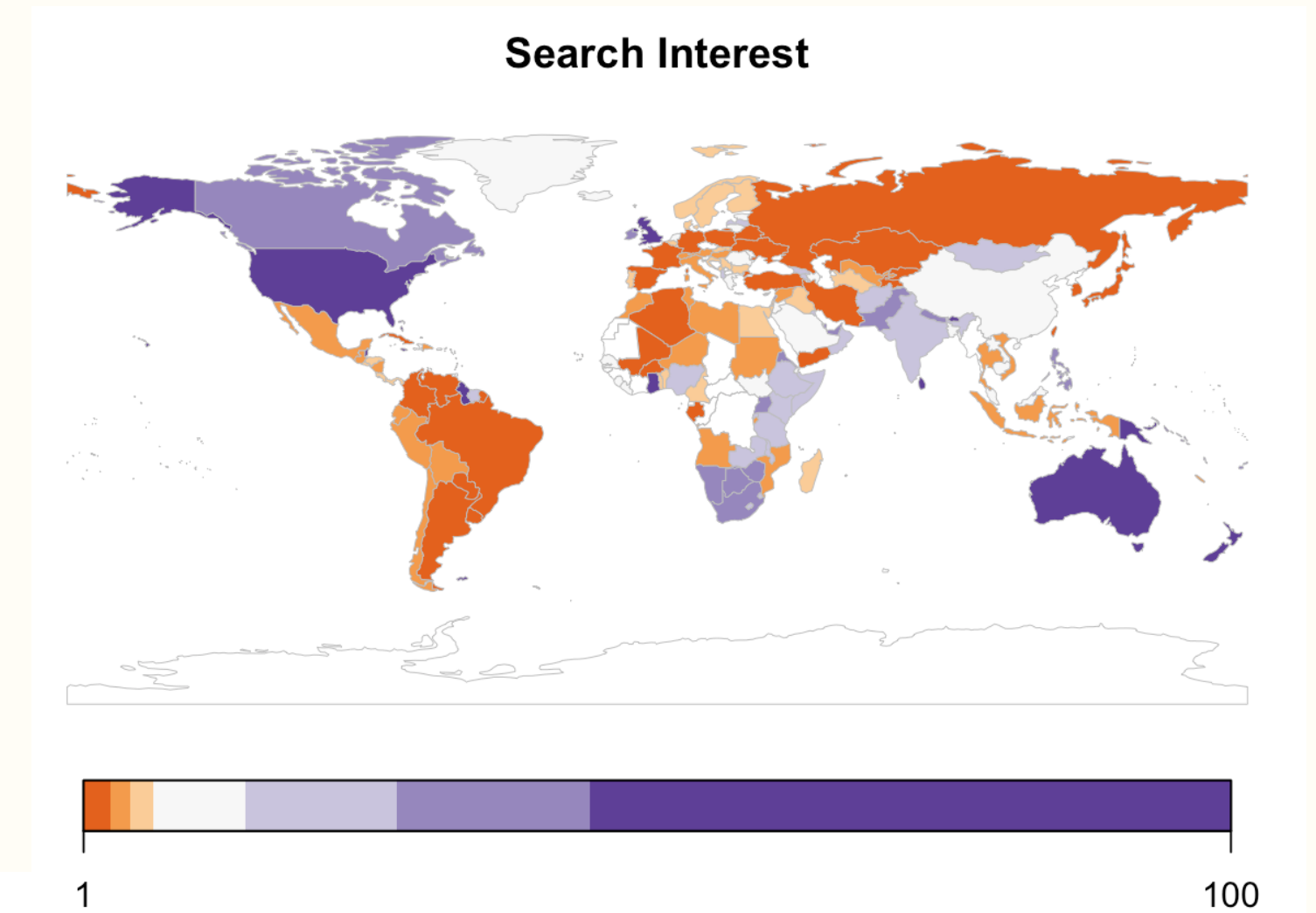
Maps

The **maps** package

Heather Mattie, PhD

Maps

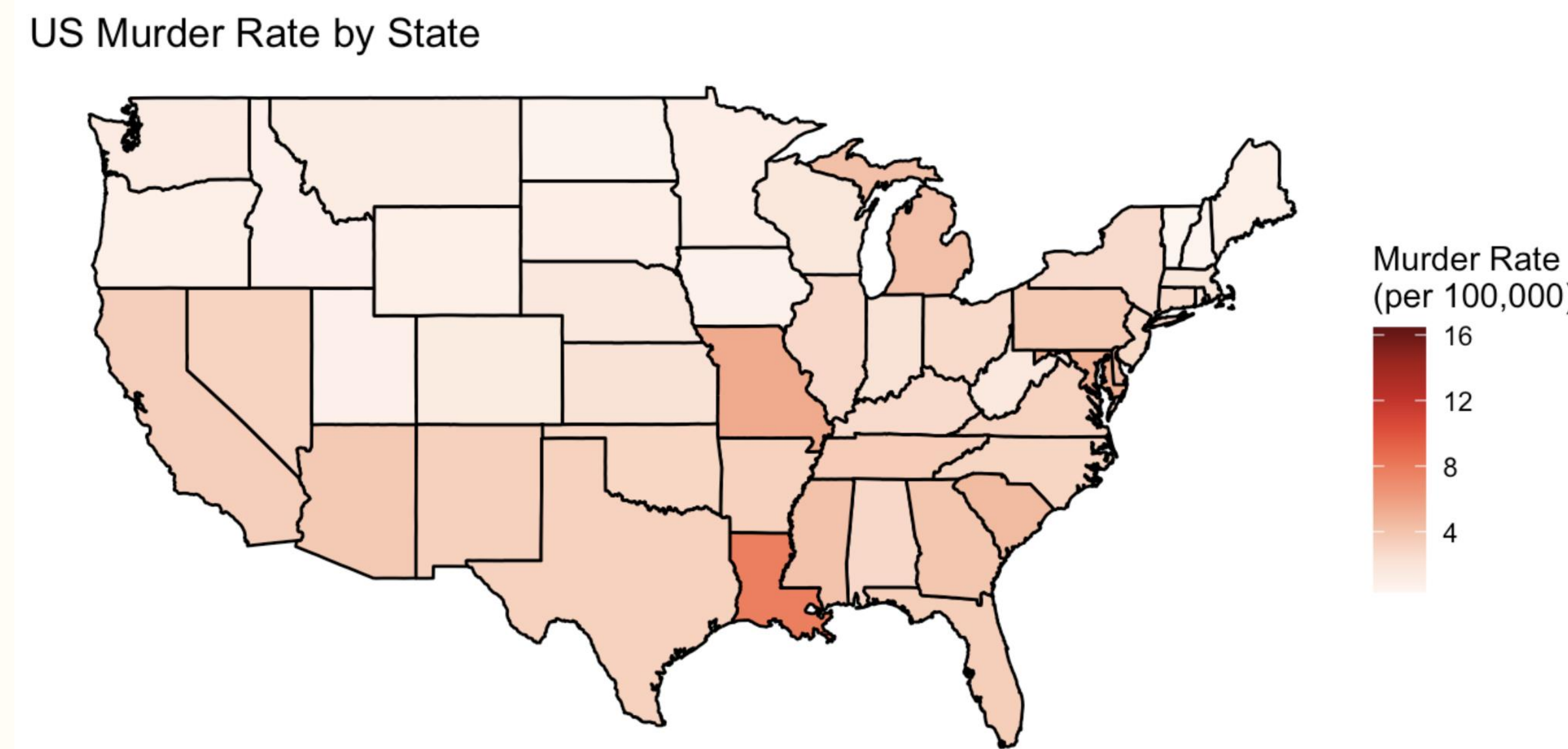
- Plots of **maps** can be very powerful, informative and aesthetically pleasing
- They can also be misleading if not created correctly
- We will use the **maps** package to create plots of maps
- Other packages: **ggspatial**, **sf**, **urbnmapr**, **rnaturalearth**, etc.



maps

The maps package contains outlines of continents, countries, states, and counties / jurisdictions / provinces

- Specify which map to use with `map_data()`
- Loaded as a data frame and can be used with the pipe (`%>%`) and `ggplot`



map_data("world")

- **long** = longitude
- **lat** = latitude
- **group** indicates which countries can be drawn together
- **order** indicates which order to connect the **lat** and **long** dots
- **region** = country
- **subregion** = county, political jurisdictions, etc.

```
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

##		long	lat	group	order	region	subregion
##	1	-69.89912	12.45200	1	1	Aruba	<NA>
##	2	-69.89571	12.42300	1	2	Aruba	<NA>
##	3	-69.94219	12.43853	1	3	Aruba	<NA>
##	4	-70.00415	12.50049	1	4	Aruba	<NA>
##	5	-70.06612	12.54697	1	5	Aruba	<NA>
##	6	-70.05088	12.59707	1	6	Aruba	<NA>

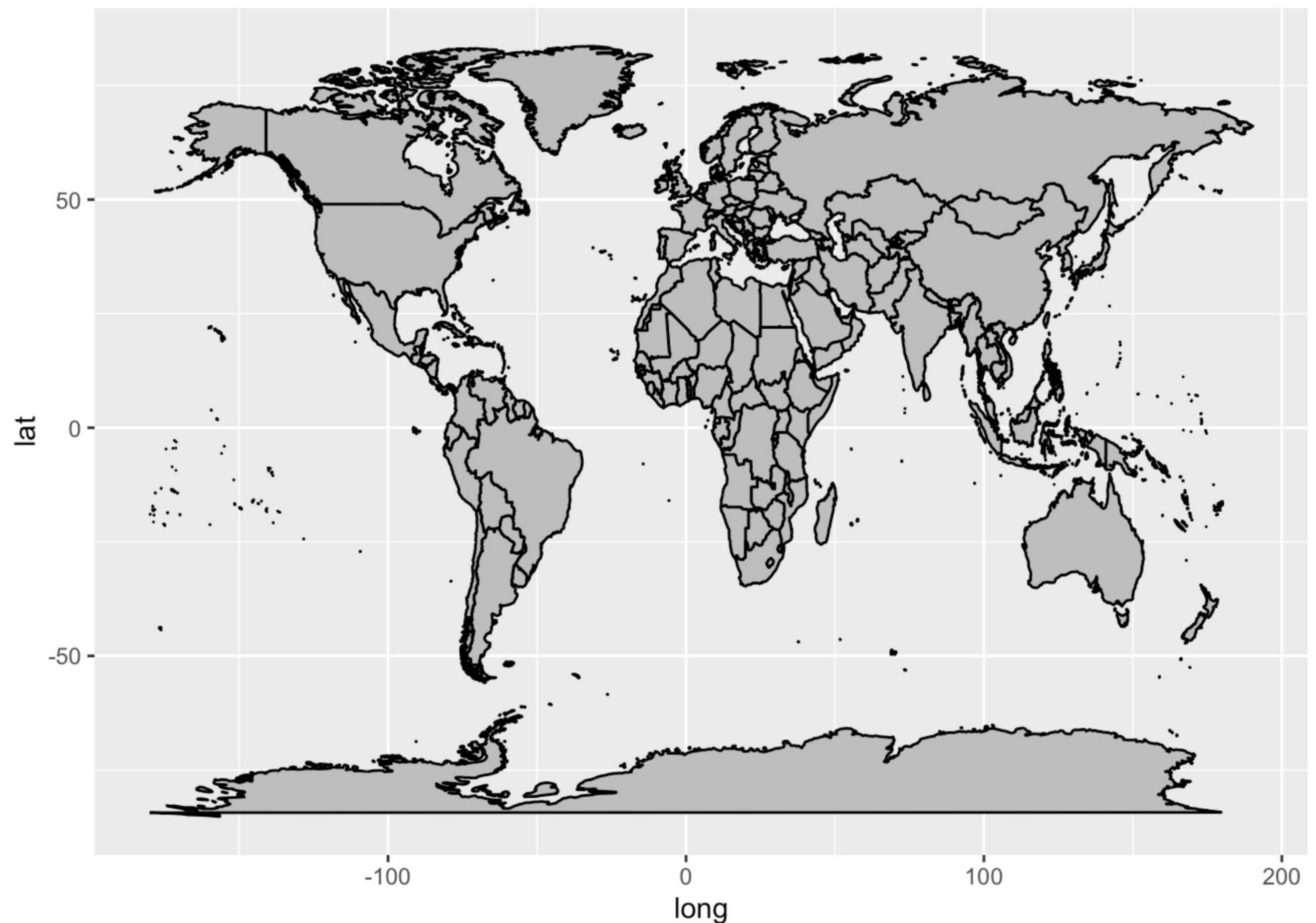
```
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```

map_data("world")

```
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

```
##           long      lat group order region subregion
## 1 -69.89912 12.45200     1     1  Aruba    <NA>
## 2 -69.89571 12.42300     1     2  Aruba    <NA>
## 3 -69.94219 12.43853     1     3  Aruba    <NA>
## 4 -70.00415 12.50049     1     4  Aruba    <NA>
## 5 -70.06612 12.54697     1     5  Aruba    <NA>
## 6 -70.05088 12.59707     1     6  Aruba    <NA>
```

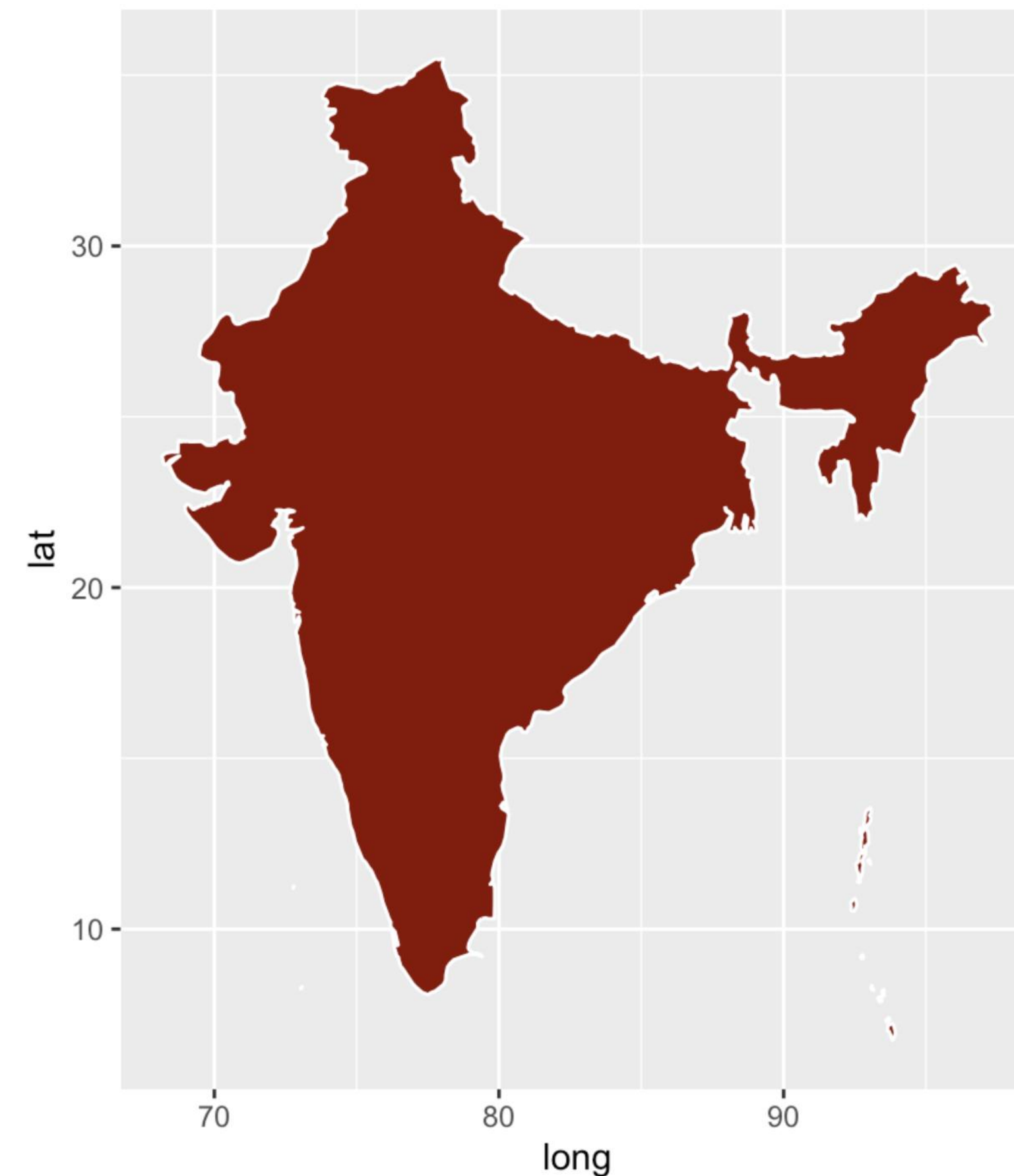
```
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```



Countries

```
india_map <- map_data("world", "India")

india_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.2)
```

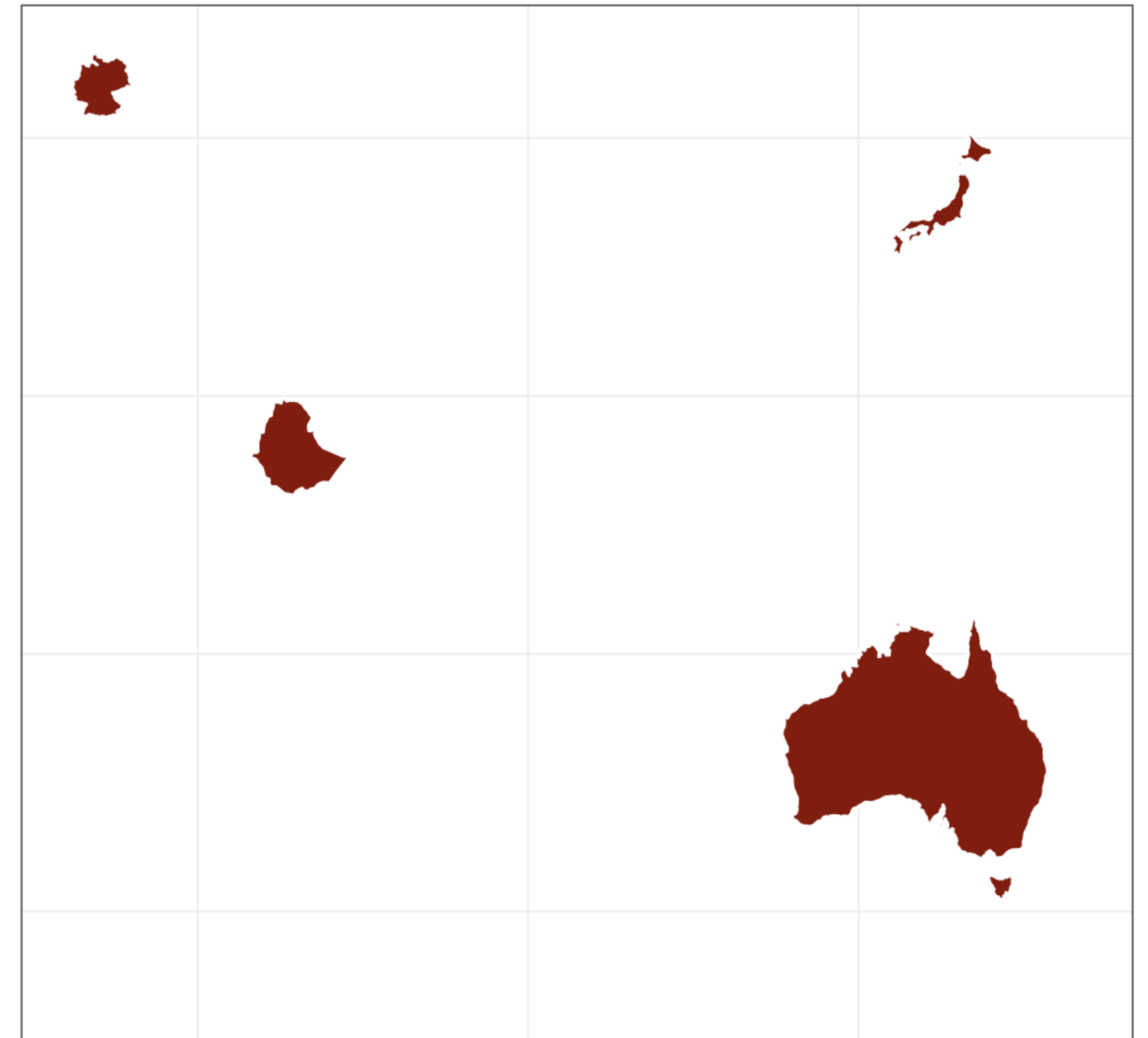


Countries

```
theme_set(theme_bw())

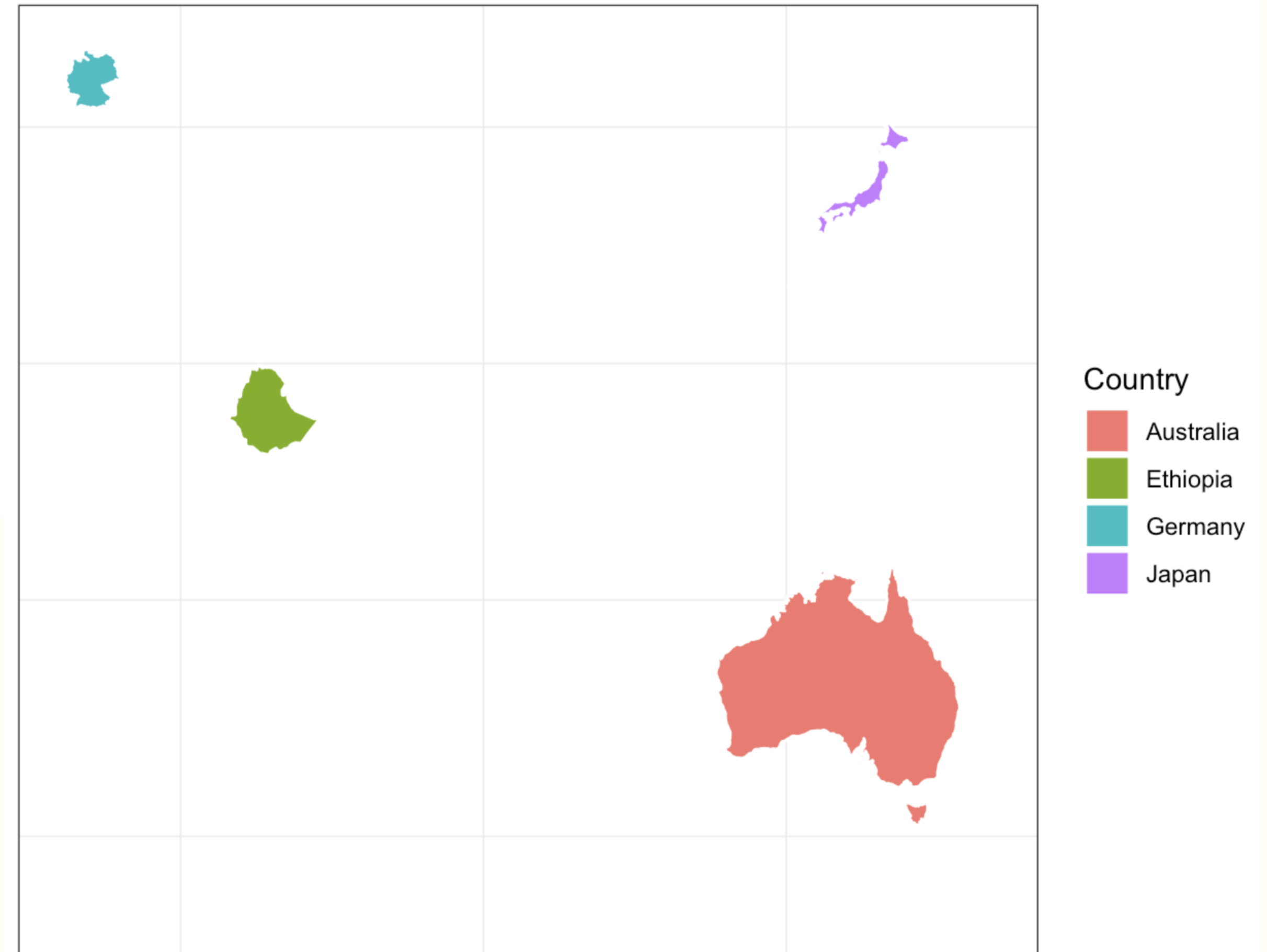
multiple_countries_map <- map_data("world",
                                   c("Japan", "Ethiopia", "Australia", "Germany"))

multiple_countries_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  coord_fixed(1.3)
```



Countries

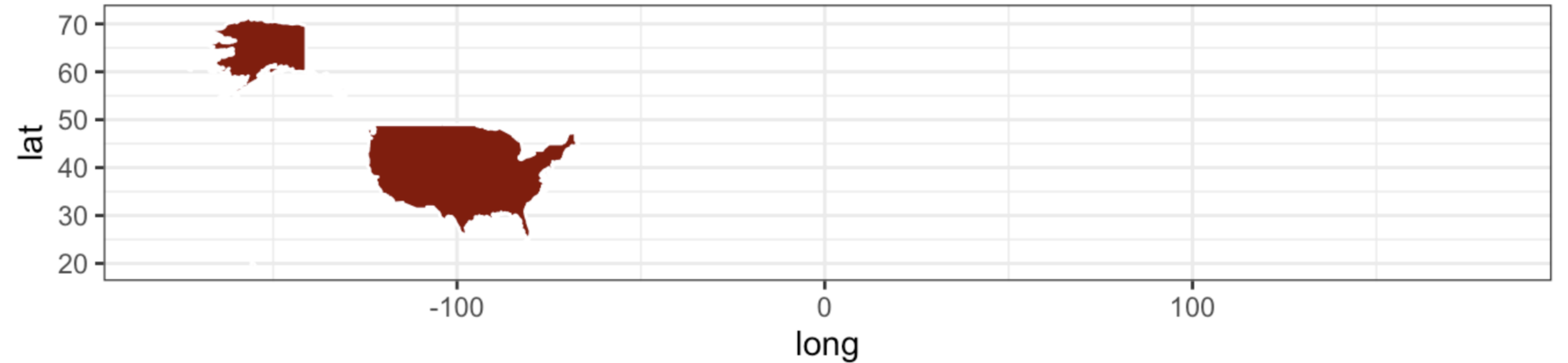
```
multiple_countries_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = region)) +  
  geom_polygon(color = "white") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  labs(fill = "Country") +  
  coord_fixed(1.3)
```



US maps

```
us_map <- map_data("world", "usa")

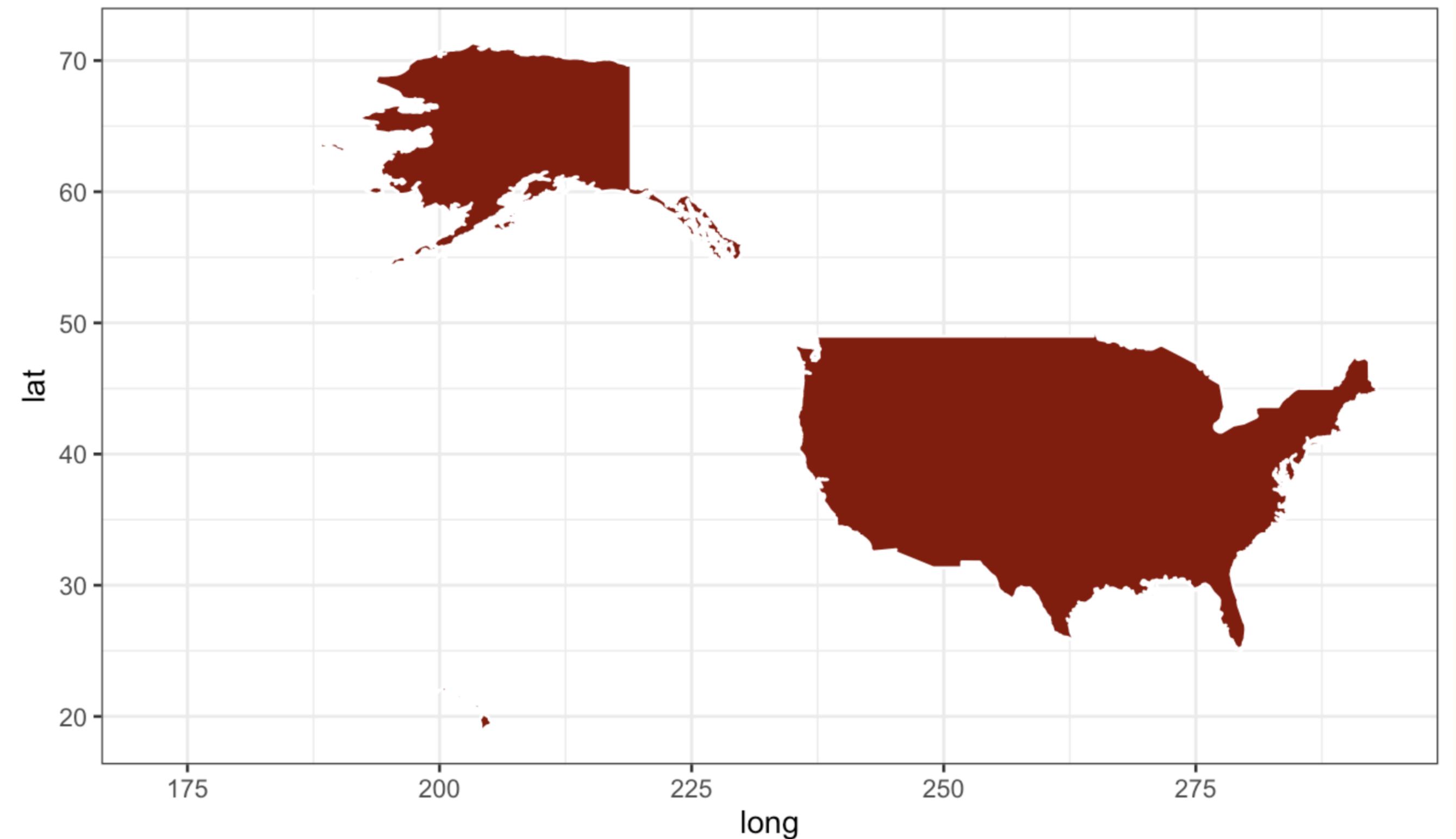
us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



US maps

```
better_us_map <- map_data("world2", "USA")

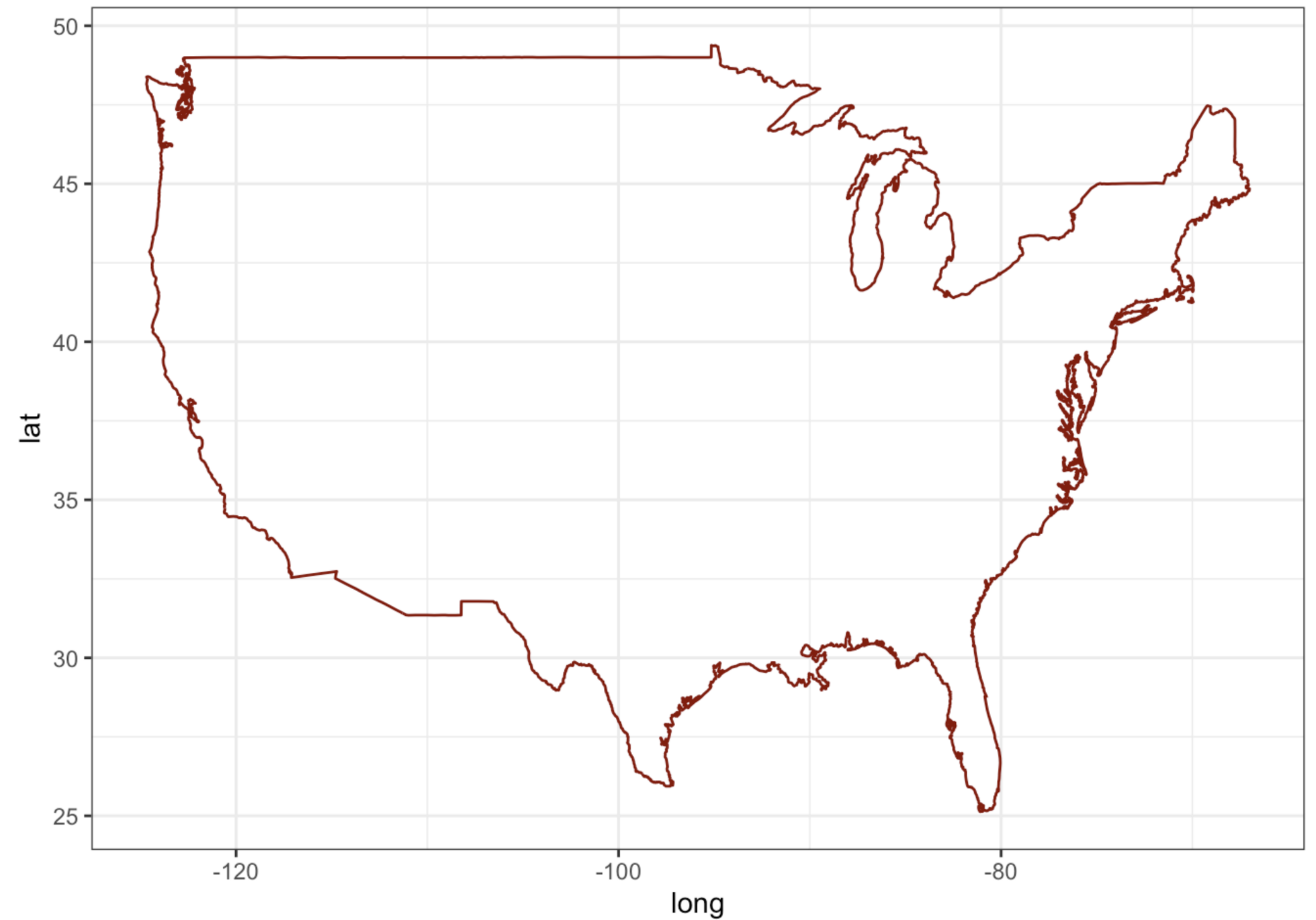
better_us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



US maps

```
usa <- map_data("usa")

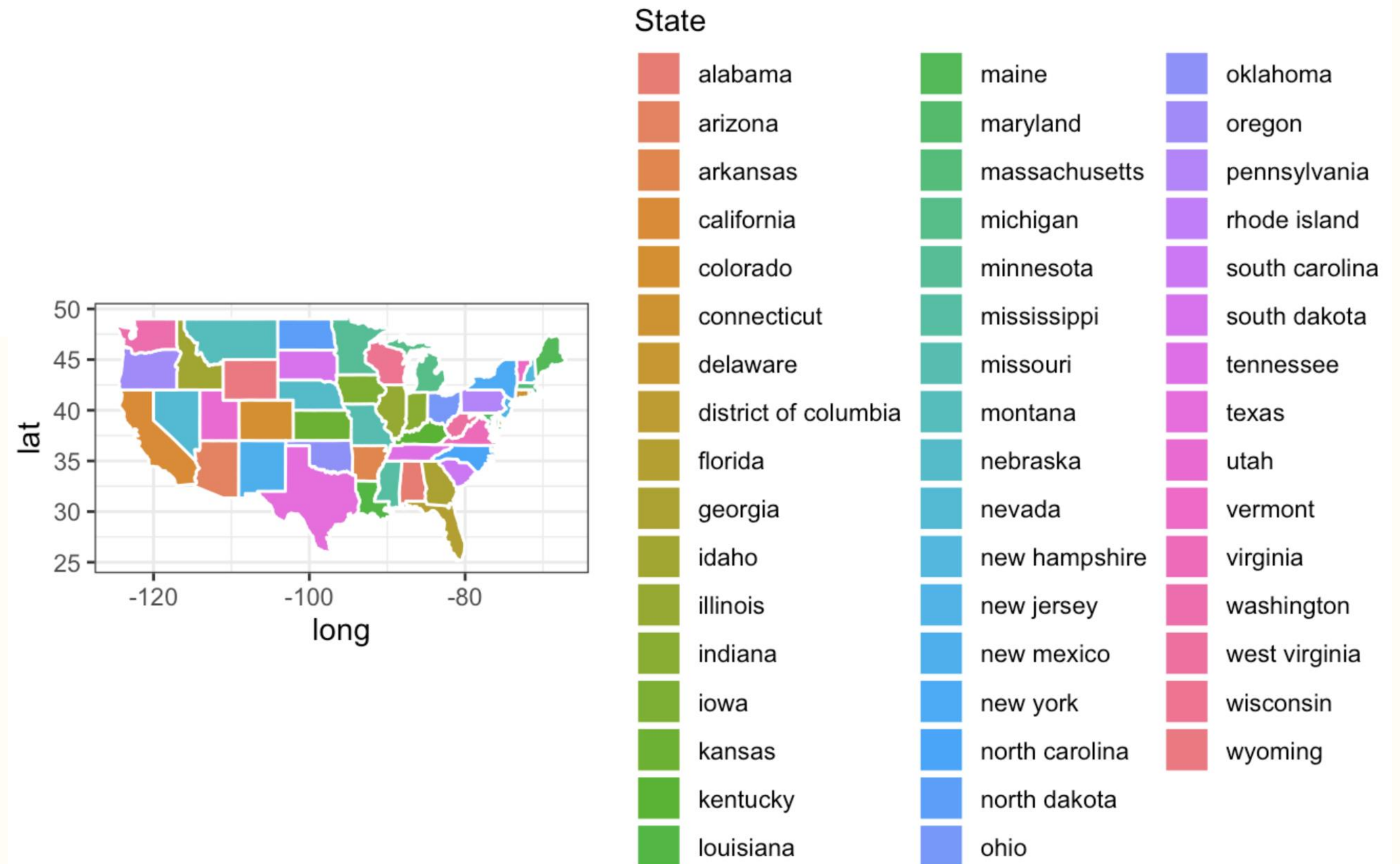
usa %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA)
```



US states

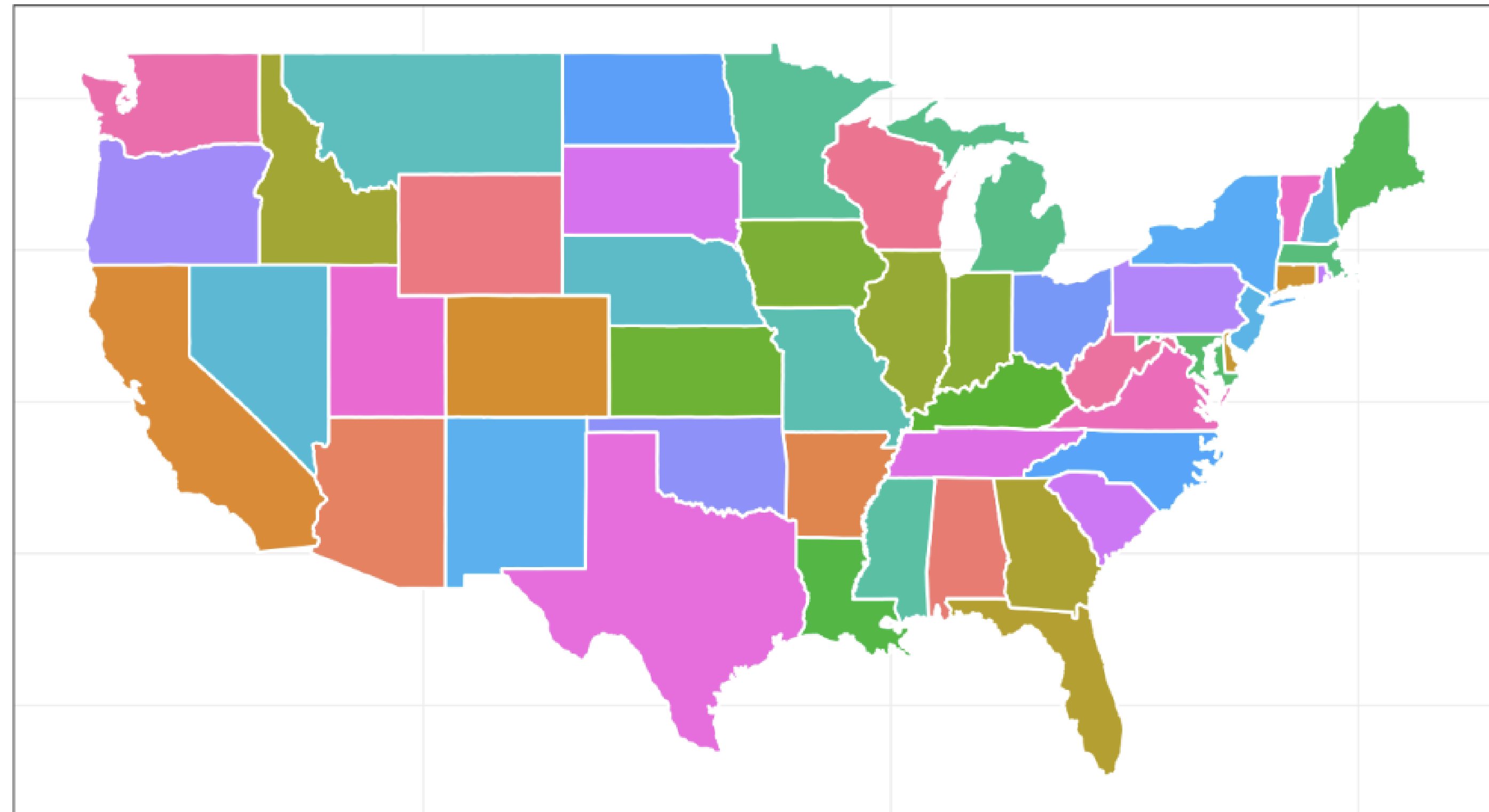
```
us_states <- map_data("state")

us_states %>%
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +
  geom_polygon(color = "white") +
  labs(fill = "State") +
  coord_fixed(1.3)
```



US states

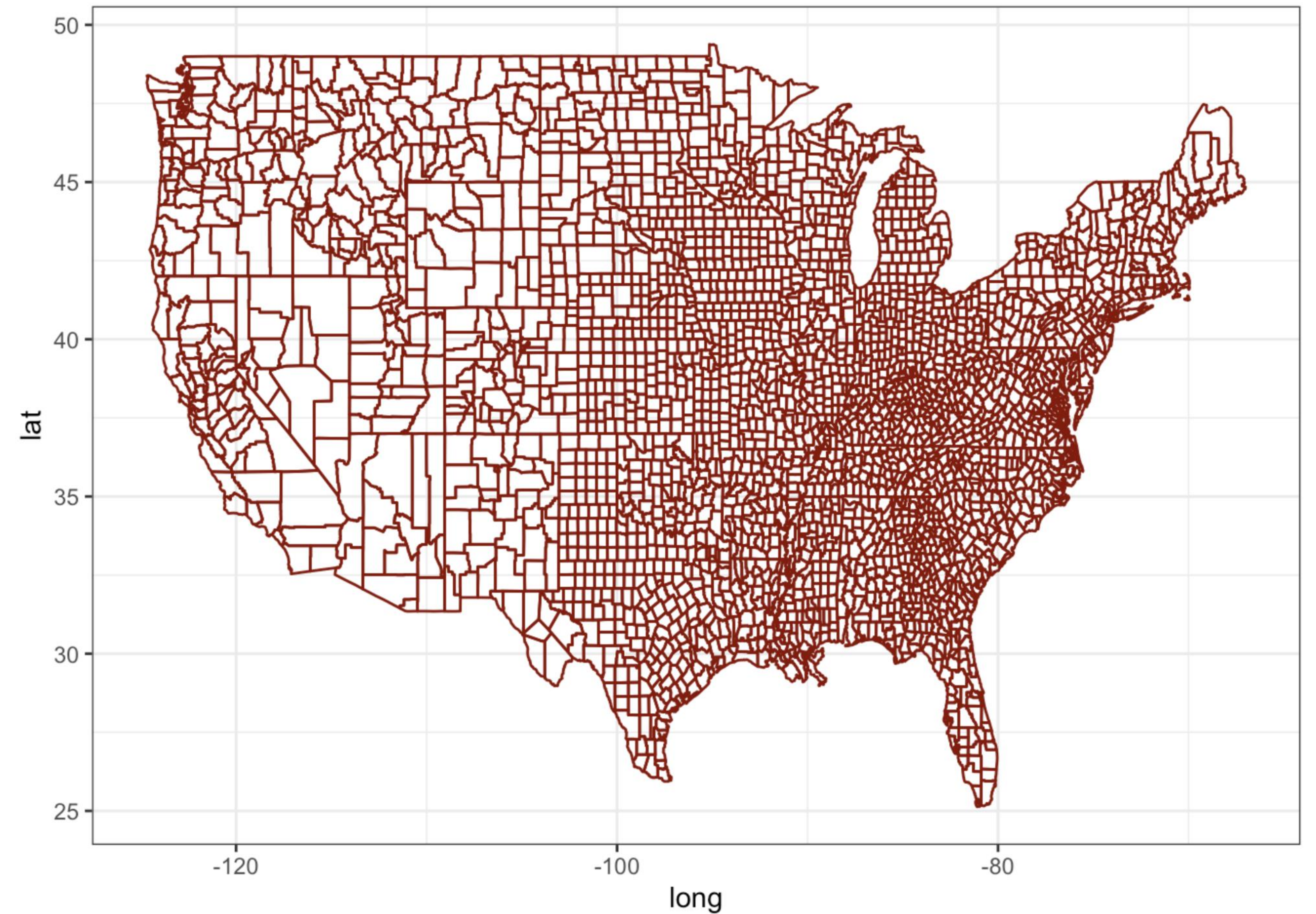
```
us_states %>%  
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +  
  geom_polygon(color = "white") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  guides(fill = "none")
```



US counties

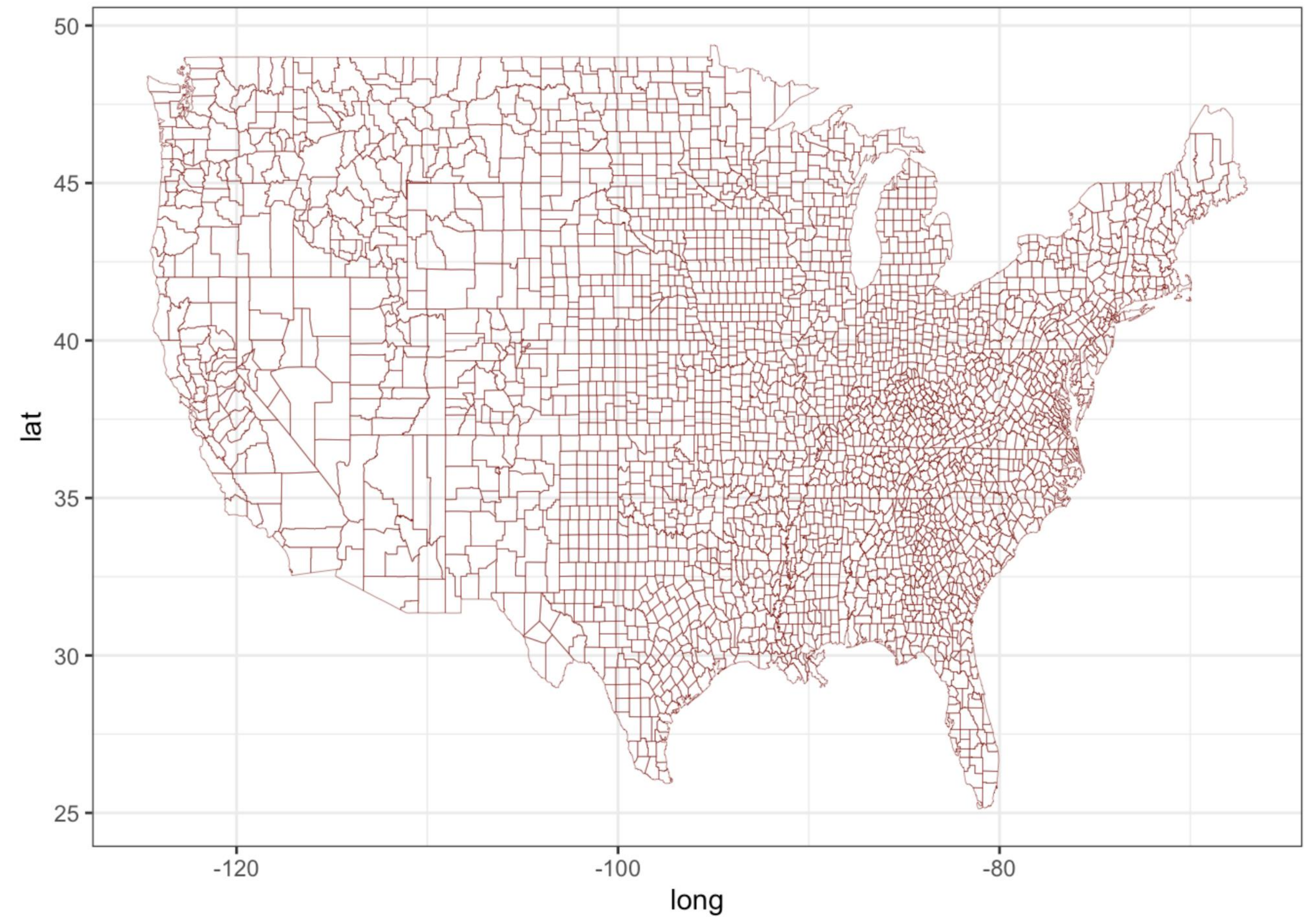
```
AllCounty <- map_data("county")
```

```
AllCounty %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(color = "darkred", fill = NA)
```



US counties

```
AllCounty %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(color = "darkred", fill = NA, linewidth = .1)
```



Summary

- The **maps** package can be used to produce maps of continents, countries, states, counties, and provinces
- The **map_data()** function loads map data
- Color or labels should be used instead of a legend if there are too many categories (countries, states, etc.) represented in the map

Maps

Plotting global life expectancy

Heather Mattie, PhD

Plotting global life expectancy

Task: plot a map displaying the life expectancy in 2015 for countries with data in the **gapminder** dataset

```
library(plyr)
library(tidyverse)
library(maps)
library(dslabs)

data(gapminder)
gapminder_2015 <- gapminder %>%
  filter(year == 2015)

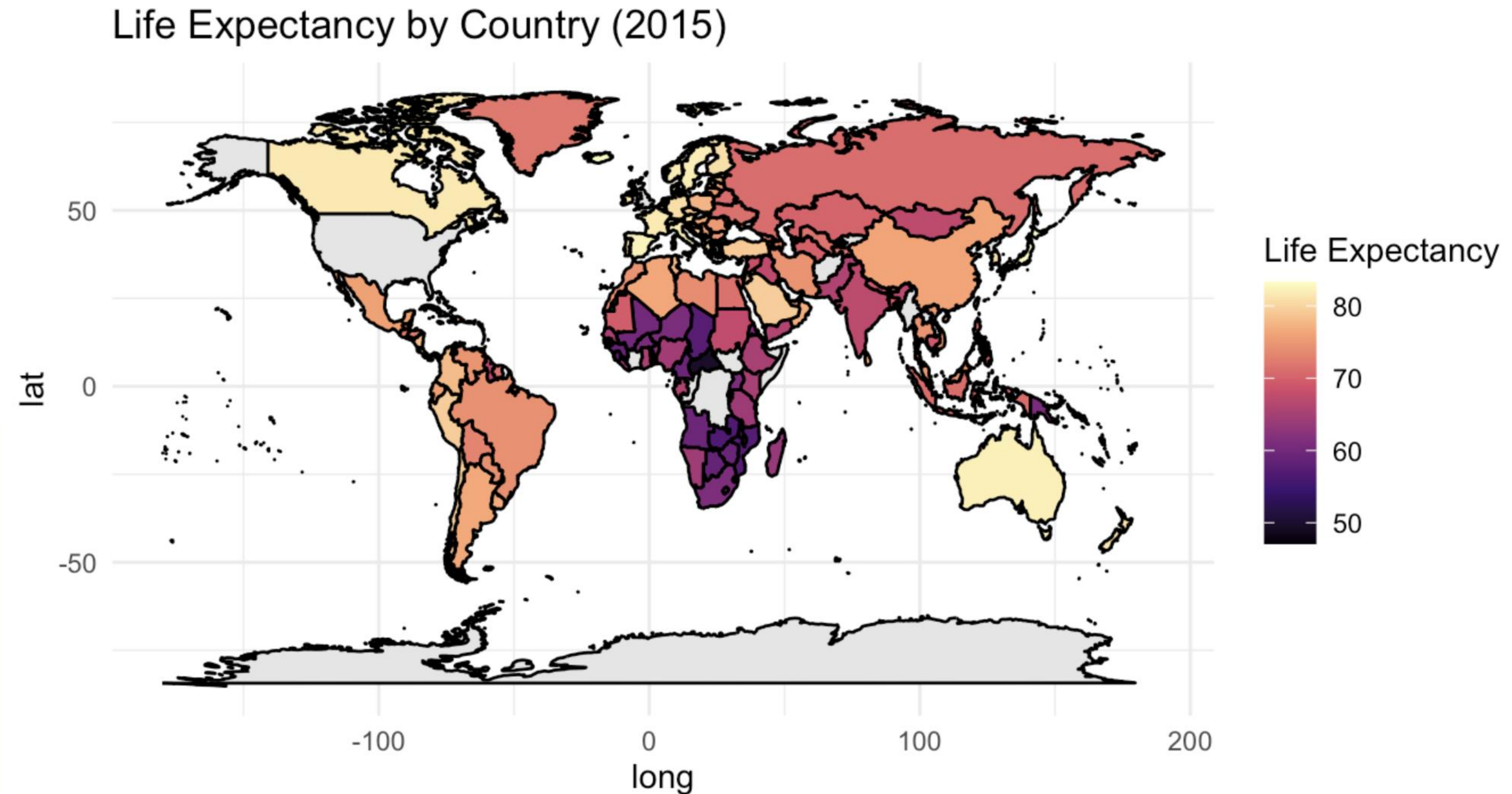
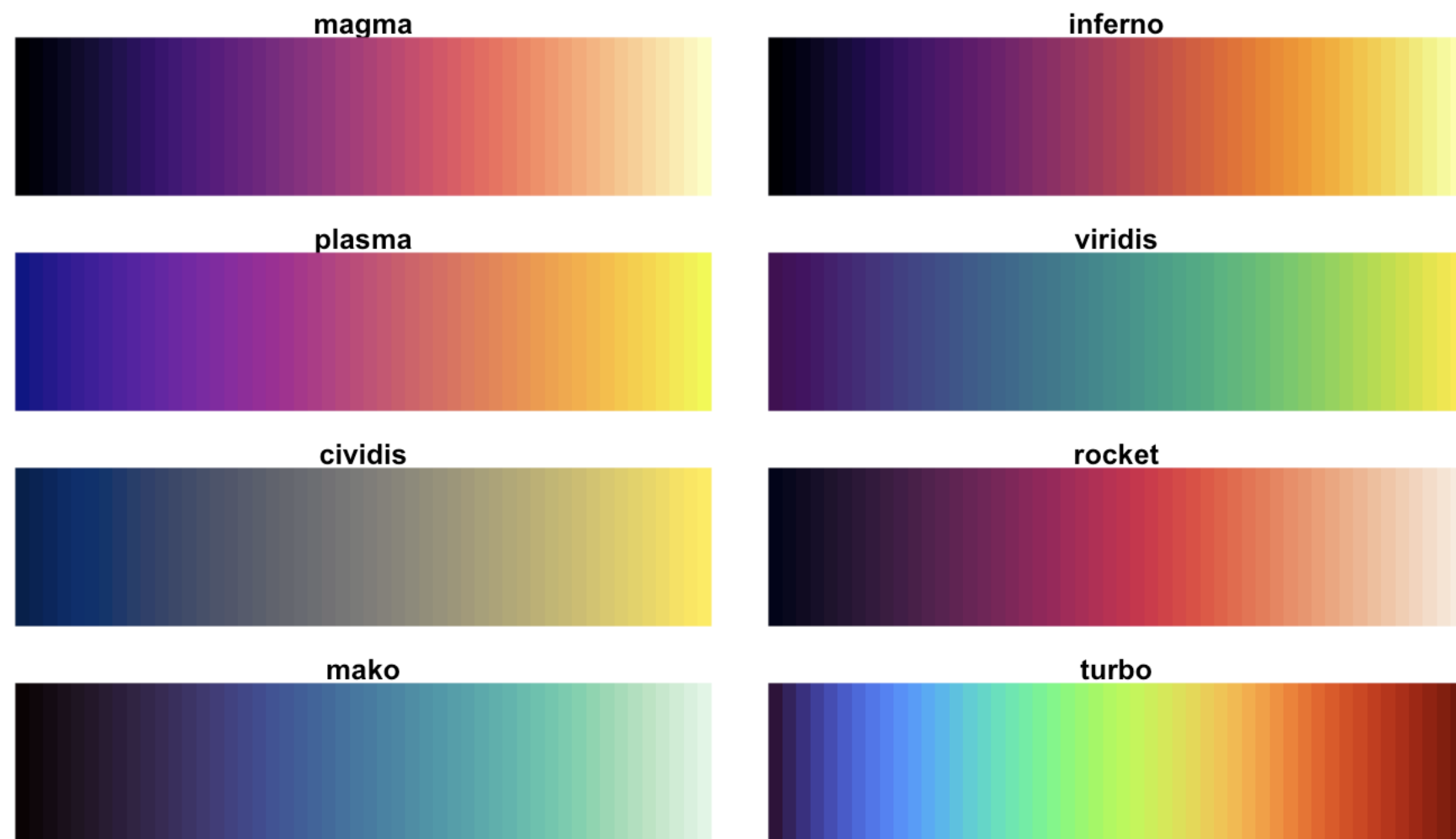
world_map <- map_data("world")
merged_world <- left_join(world_map, gapminder_2015,
                          by = c("region" = "country"))

head(merged_world)
```

```
##      long      lat group order region subregion year infant_mortality
## 1 -69.89912 12.45200     1     1  Aruba      <NA> 2015              NA
## 2 -69.89571 12.42300     1     2  Aruba      <NA> 2015              NA
## 3 -69.94219 12.43853     1     3  Aruba      <NA> 2015              NA
## 4 -70.00415 12.50049     1     4  Aruba      <NA> 2015              NA
## 5 -70.06612 12.54697     1     5  Aruba      <NA> 2015              NA
## 6 -70.05088 12.59707     1     6  Aruba      <NA> 2015              NA
##      life_expectancy fertility population gdp continent region.y
## 1             75.72      1.66     103889   NA  Americas Caribbean
## 2             75.72      1.66     103889   NA  Americas Caribbean
## 3             75.72      1.66     103889   NA  Americas Caribbean
## 4             75.72      1.66     103889   NA  Americas Caribbean
## 5             75.72      1.66     103889   NA  Americas Caribbean
## 6             75.72      1.66     103889   NA  Americas Caribbean
```

Plotting global life expectancy

```
ggplot(merged_world, aes(long, lat, group = group,  
                          fill = life_expectancy)) +  
  geom_polygon(color = "black") +  
  scale_fill_viridis_c(option = "magma",  
                       na.value = "gray90") +  
  theme_minimal() +  
  labs(title = "Life Expectancy by Country (2015)",  
       fill = "Life Expectancy") +  
  coord_fixed(1.3)
```



Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
## [1] "Afghanistan"      "American Samoa"
## [3] "Andorra"          "Anguilla"
## [5] "Antarctica"       "Antigua"
## [7] "Ascension Island" "Azores"
## [9] "Barbuda"          "Bermuda"
## [11] "Bonaire"          "Canary Islands"
## [13] "Cayman Islands"   "Chagos Archipelago"
## [15] "Christmas Island" "Cocos Islands"
## [17] "Cook Islands"     "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands" "Faroe Islands"
## [23] "French Guiana"    "French Southern and Antarctic Lands"
## [25] "Grenadines"       "Guadeloupe"
## [27] "Guam"             "Guernsey"
## [29] "Heard Island"     "Isle of Man"
## [31] "Ivory Coast"      "Jersey"
## [33] "Kosovo"           "Kyrgyzstan"
## [35] "Laos"             "Liechtenstein"
## [37] "Madeira Islands"  "Marshall Islands"
## [39] "Martinique"       "Mayotte"
## [41] "Micronesia"       "Monaco"
## [43] "Montserrat"       "Myanmar"
## [45] "Nauru"            "Nevis"
```

```
levels(gapminder_2015$country)
```

```
## [1] "Albania"          "Algeria"
## [3] "Angola"           "Antigua and Barbuda"
## [5] "Argentina"        "Armenia"
## [7] "Aruba"            "Australia"
## [9] "Austria"          "Azerbaijan"
## [11] "Bahamas"          "Bahrain"
## [13] "Bangladesh"       "Barbados"
## [15] "Belarus"          "Belgium"
## [17] "Belize"           "Benin"
## [19] "Bhutan"           "Bolivia"
## [21] "Bosnia and Herzegovina" "Botswana"
## [23] "Brazil"           "Brunei"
## [25] "Bulgaria"         "Burkina Faso"
## [27] "Burundi"          "Cambodia"
## [29] "Cameroon"         "Canada"
## [31] "Cape Verde"       "Central African Republic"
## [33] "Chad"             "Chile"
## [35] "China"            "Colombia"
## [37] "Comoros"          "Congo, Dem. Rep."
## [39] "Congo, Rep."      "Costa Rica"
## [41] "Cote d'Ivoire"    "Croatia"
## [43] "Cuba"             "Cyprus"
## [45] "Czech Republic"  "Denmark"
## [47] "Djibouti"         "Dominican Republic"
## [49] "Ecuador"          "Egypt"
## [51] "El Salvador"      "Equatorial Guinea"
## [53] "Eritrea"          "Estonia"
```



Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
## [1] "Afghanistan"      "American Samoa"
## [3] "Andorra"          "Anguilla"
## [5] "Antarctica"       "Antigua"
## [7] "Ascension Island" "Azores"
## [9] "Barbuda"          "Bermuda"
## [11] "Bonaire"          "Canary Islands"
## [13] "Cayman Islands"   "Chagos Archipelago"
## [15] "Christmas Island" "Cocos Islands"
## [17] "Cook Islands"     "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands" "Faroe Islands"
## [23] "French Guiana"    "French Southern and Antarctic Lands"
## [25] "Grenadines"       "Guadeloupe"
## [27] "Guam"             "Guernsey"
## [29] "Heard Island"     "Isle of Man"
## [31] "Ivory Coast"      "Jersey"
## [33] "Kosovo"           "Kyrgyzstan"
## [35] "Laos"             "Liechtenstein"
## [37] "Madeira Islands"  "Marshall Islands"
## [39] "Martinique"       "Mayotte"
## [41] "Micronesia"       "Monaco"
## [43] "Montserrat"       "Myanmar"
## [45] "Nauru"            "Nevis"
```

```
levels(gapminder_2015$country)
```

```
## [1] "Albania"          "Algeria"
## [3] "Angola"           "Antigua and Barbuda"
## [5] "Argentina"        "Armenia"
## [7] "Aruba"            "Australia"
## [9] "Austria"          "Azerbaijan"
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## [17] "Belize"           "Benin"
## [19] "Bhutan"           "Bolivia"
## [21] "Bosnia and Herzegovina" "Botswana"
## [23] "Brazil"           "Brunei"
## [25] "Bulgaria"         "Burkina Faso"
## [27] "Burundi"          "Cambodia"
## [29] "Cameroon"         "Canada"
## [31] "Cape Verde"       "Central African Republic"
## [33] "Chad"             "Chile"
## [35] "China"            "Colombia"
## [37] "Comoros"          "Congo, Dem. Rep."
## [39] "Congo, Rep."      "Costa Rica"
## [41] "Cote d'Ivoire"    "Croatia"
## [43] "Cuba"             "Cyprus"
## [45] "Czech Republic"  "Denmark"
## [47] "Djibouti"         "Dominican Republic"
## [49] "Ecuador"          "Egypt"
## [51] "El Salvador"      "Equatorial Guinea"
## [53] "Eritrea"          "Estonia"
```



Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
## [1] "Afghanistan"      "American Samoa"
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## [5] "Antarctica"       "Antigua"
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## [11] "Bonaire"          "Canary Islands"
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## [15] "Christmas Island" "Cocos Islands"
## [17] "Cook Islands"     "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
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## [27] "Guam"             "Guernsey"
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## [35] "Laos"             "Liechtenstein"
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## [41] "Micronesia"       "Monaco"
## [43] "Montserrat"       "Myanmar"
## [45] "Nauru"            "Nevis"
```

```
levels(gapminder_2015$country)
```

```
## [1] "Albania"      "Algeria"
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## [5] "Argentina"    "Armenia"
## [7] "Aruba"        "Australia"
## [9] "Austria"      "Azerbaijan"
## [11] "Bahamas"      "Bahrain"
## [13] "Bangladesh"   "Barbados"
## [15] "Belarus"      "Belgium"
## [17] "Belize"       "Benin"
## [19] "Bhutan"       "Bolivia"
## [21] "Bosnia and Herzegovina" "Botswana"
## [23] "Brazil"       "Brunei"
## [25] "Bulgaria"     "Burkina Faso"
## [27] "Burundi"     "Cambodia"
## [29] "Cameroon"     "Canada"
## [31] "Cape Verde"   "Central African Republic"
## [33] "Chad"         "Chile"
## [35] "China"        "Colombia"
## [37] "Comoros"      "Congo, Dem. Rep."
## [39] "Congo, Rep."  "Costa Rica"
## [41] "Cote d'Ivoire" "Croatia"
## [43] "Cuba"         "Cyprus"
## [45] "Czech Republic" "Denmark"
## [47] "Djibouti"     "Dominican Republic"
## [49] "Ecuador"      "Egypt"
## [51] "El Salvador"  "Equatorial Guinea"
## [53] "Eritrea"      "Estonia"
```

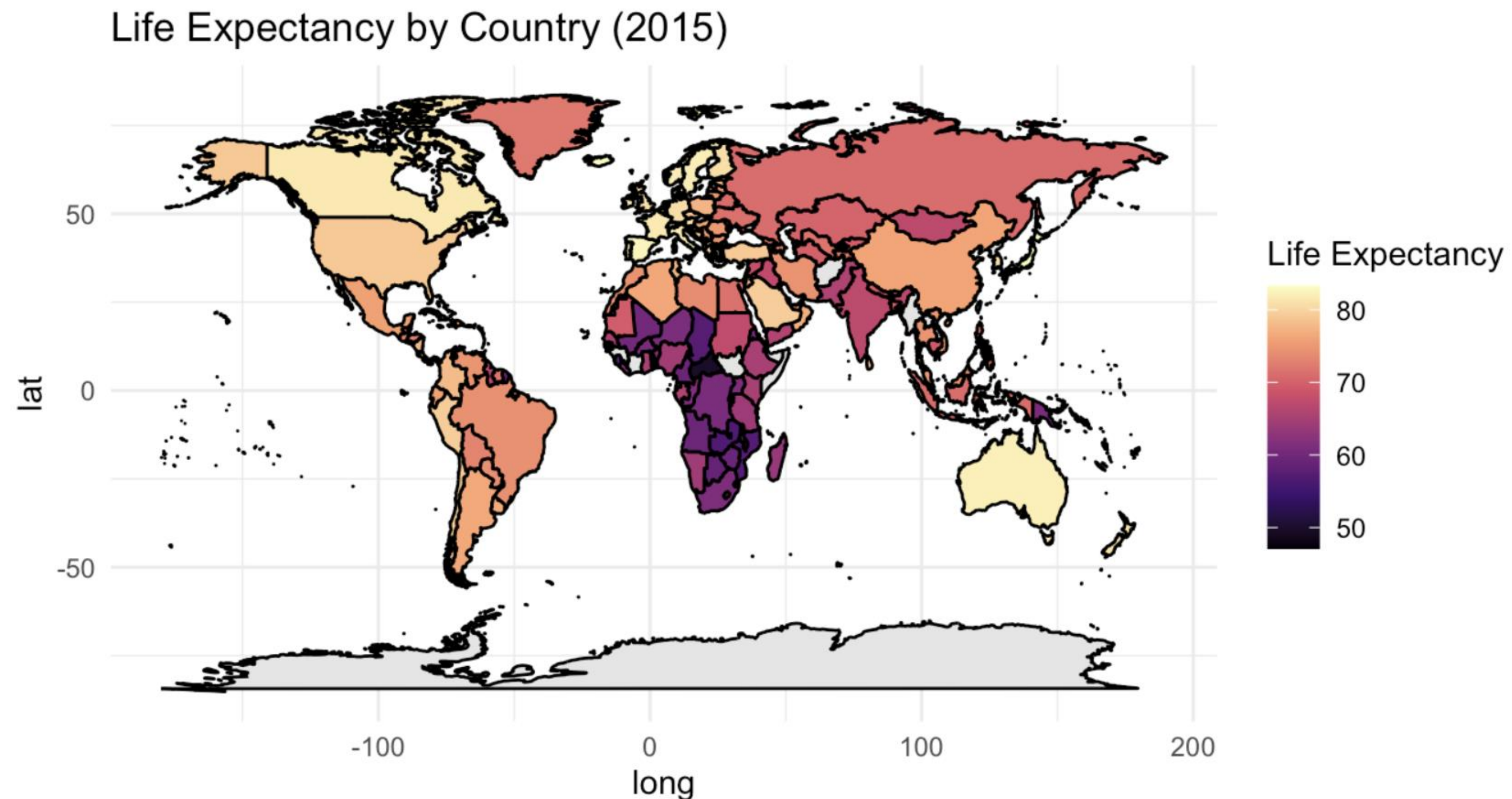


Renaming

```
gapminder_2015 <- gapminder %>% filter(year == 2015) %>%  
  mutate(country =  
    revalue(country, c("Antigua and Barbuda" = "Antigua",  
                        "Samoa" = "American Samoa",  
                        "Congo, Rep." = "Republic of Congo",  
                        "Congo, Dem. Rep." = "Democratic Republic of the Congo",  
                        "Guinea" = "French Guiana",  
                        "Kyrgyz Republic" = "Kyrgyzstan",  
                        "St. Lucia" = "Saint Lucia",  
                        "St. Vincent and the Grenadines" = "Saint Vincent",  
                        "Trinidad and Tobago" = "Trinidad",  
                        "United Kingdom" = "UK",  
                        "United States" = "USA"))))
```

```
world_map <- map_data("world")  
merged_world <- left_join(world_map, gapminder_2015, by = c("region" = "country"))
```

```
ggplot(merged_world, aes(long, lat, group = group, fill = life_expectancy)) +  
  geom_polygon(color = "black") +  
  scale_fill_viridis_c(option = "magma", na.value = "gray90") +  
  theme_minimal() +  
  labs(title = "Life Expectancy by Country (2015)", fill = "Life Expectancy") +  
  coord_fixed(1.3)
```



Summary

- Inconsistencies in the spelling of locations can lead to missing data in plots of maps
- Renaming the locations in one data frame before joining to a second data frame can solve a lot of inconsistencies
- Sometimes new data will need to be collected to complete a map plot

Maps

Plotting US state murder rates

Heather Mattie, PhD

Plotting state murder rates

- Task: plot a map of the US states and fill in each state with a color representing the murder rate in that state using the **murders** dataset
- Need to use state map data and add the murder rate for each state to the map data frame

```
library(tidyverse)
library(maps)
library(dslabs)
library(RColorBrewer)

data(murders)
murders <- murders %>% mutate(rate = total / population * 100000)
states_map <- map_data("state")

head(murders)
```

```
##      state abb region population total   rate
## 1  Alabama  AL  South   4779736   135 2.824424
## 2   Alaska  AK   West    710231    19 2.675186
## 3   Arizona AZ   West   6392017   232 3.629527
## 4  Arkansas AR  South   2915918    93 3.189390
## 5 California CA   West  37253956  1257 3.374138
## 6   Colorado CO   West   5029196    65 1.292453
```

```
head(states_map)
```

```
##      long      lat group order  region subregion
## 1 -87.46201 30.38968     1     1 alabama    <NA>
## 2 -87.48493 30.37249     1     2 alabama    <NA>
## 3 -87.52503 30.37249     1     3 alabama    <NA>
## 4 -87.53076 30.33239     1     4 alabama    <NA>
## 5 -87.57087 30.32665     1     5 alabama    <NA>
## 6 -87.58806 30.32665     1     6 alabama    <NA>
```

Plotting state murder rates

- Task: plot a map of the US states and fill in each state with a color representing the murder rate in that state using the **murders** dataset
- Need to use state map data and add the murder rate for each state to the map data frame

```
murders <- murders %>% mutate(region = tolower(state))  
head(murders)
```

```
##      state abb  region population total    rate  
## 1  Alabama AL   alabama  4779736  135 2.824424  
## 2  Alaska AK    alaska   710231   19 2.675186  
## 3  Arizona AZ   arizona  6392017  232 3.629527  
## 4  Arkansas AR  arkansas  2915918   93 3.189390  
## 5 California CA california 37253956 1257 3.374138  
## 6  Colorado CO   colorado  5029196   65 1.292453
```

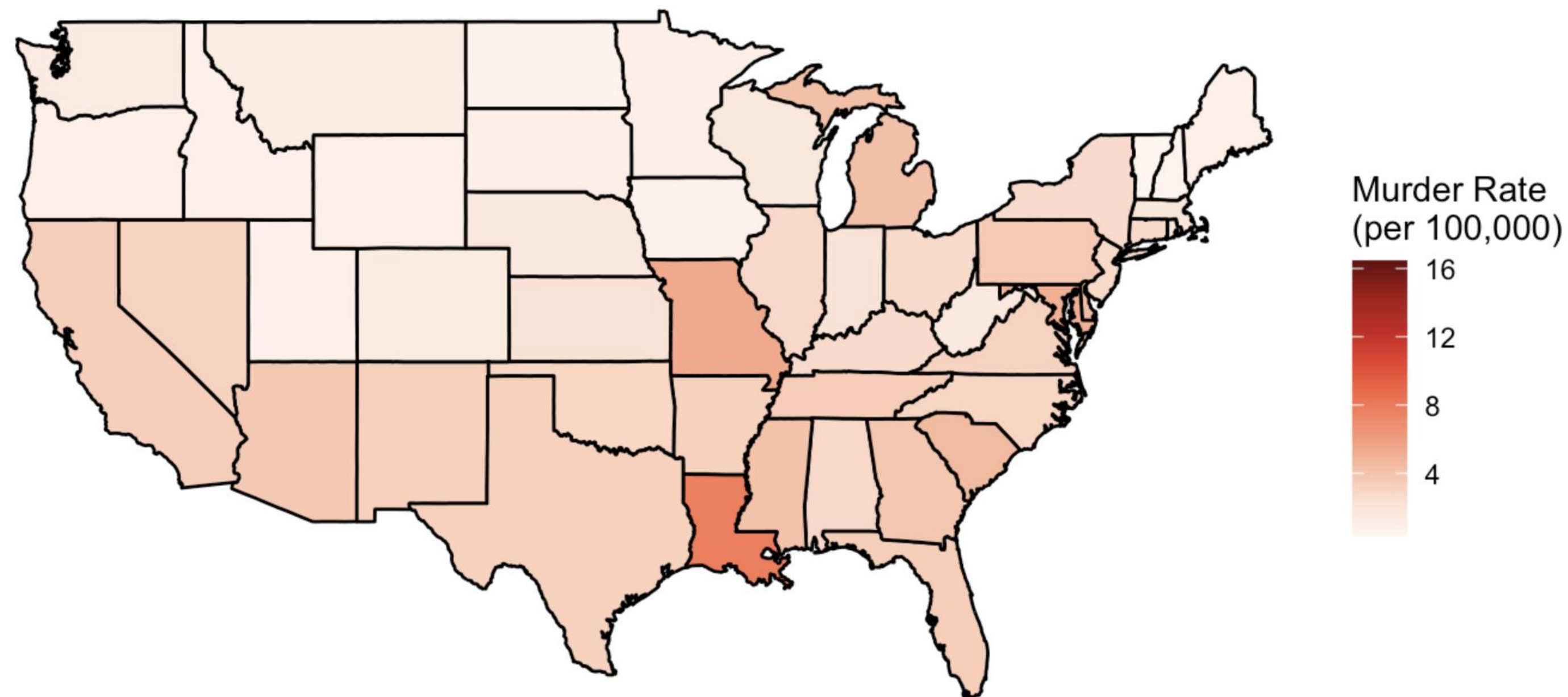
```
merged_data <- left_join(states_map, murders, by = "region")  
head(merged_data)
```

```
##      long    lat group order region subregion state abb population total  
## 1 -87.46201 30.38968     1     1 alabama      <NA> Alabama AL   4779736  135  
## 2 -87.48493 30.37249     1     2 alabama      <NA> Alabama AL   4779736  135  
## 3 -87.52503 30.37249     1     3 alabama      <NA> Alabama AL   4779736  135  
## 4 -87.53076 30.33239     1     4 alabama      <NA> Alabama AL   4779736  135  
## 5 -87.57087 30.32665     1     5 alabama      <NA> Alabama AL   4779736  135  
## 6 -87.58806 30.32665     1     6 alabama      <NA> Alabama AL   4779736  135  
##      rate  
## 1 2.824424  
## 2 2.824424  
## 3 2.824424  
## 4 2.824424  
## 5 2.824424  
## 6 2.824424
```

Plotting state murder rates

```
merged_data %>%  
  ggplot(aes(long, lat, group = group, fill = rate)) +  
    geom_polygon(color = "black") +  
    scale_fill_gradientn(colors = brewer.pal(9, "Reds")) +  
    theme_void() +  
    labs(title = "US Murder Rate by State",  
         fill = "Murder Rate \n(per 100,000)") +  
    coord_fixed(1.3)
```

US Murder Rate by State



Adding state labels

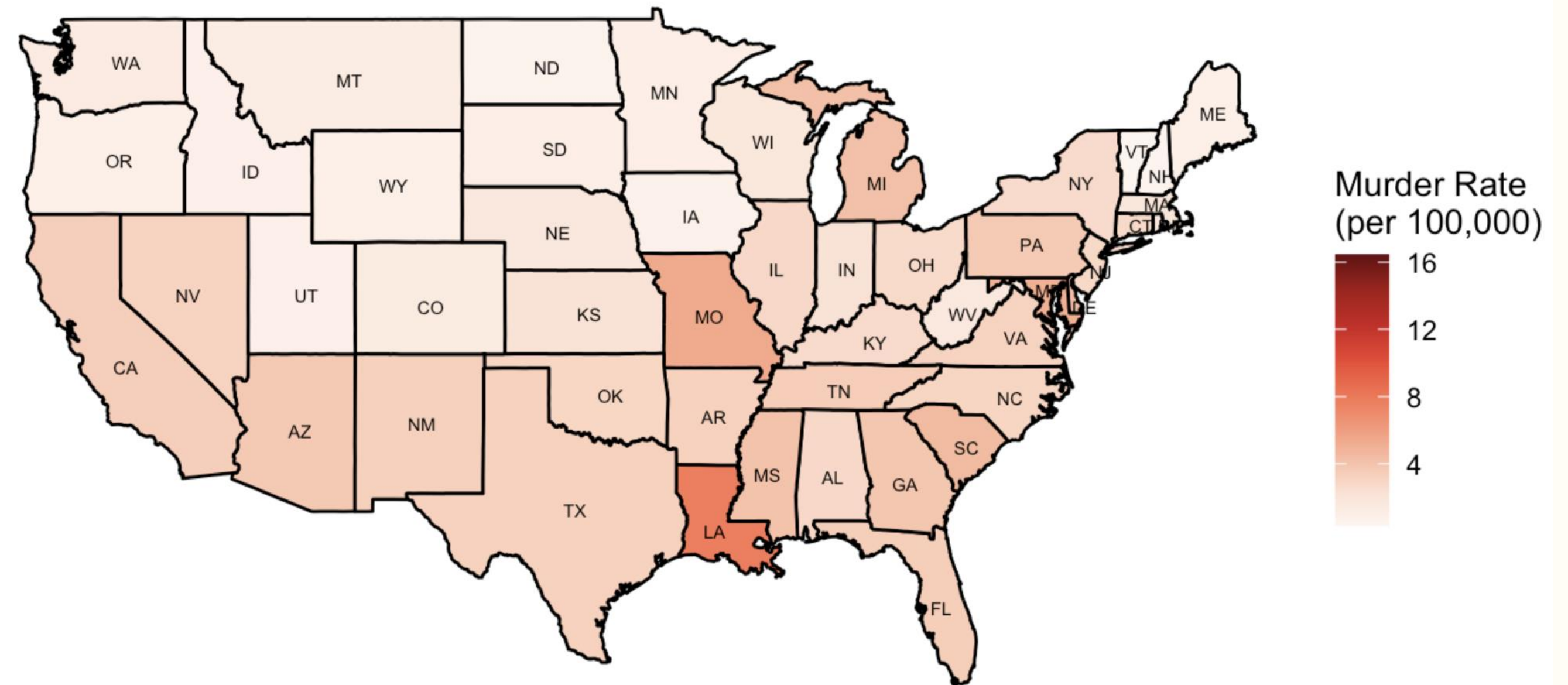
```
library(ggrepel)
centroids <- data.frame(region = tolower(state.name),
                        long = state.center$x,
                        lat = state.center$y)

centroids$abb <- state.abb[match(centroids$region,
                                tolower(state.name))]

centroids <- centroids %>%
  filter(!region %in% c("alaska", "hawaii"))

merged_data %>%
  ggplot(aes(long, lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  scale_fill_gradientn(colors = brewer.pal(9, "Reds")) +
  theme_void() +
  labs(title = "US Murder Rate by State",
       fill = "Murder Rate \n(per 100,000)") +
  coord_fixed(1.3) +
  with(centroids,
       annotate(geom = "text", x = long, y = lat, label = abb,
               size = 2, color = "black", family = "Arial"))
```

US Murder Rate by State



Summary

- `map_data("states")` can be used to create a map of the contiguous United States
- The `tolower` function makes all letters in a string lowercase
- `state.center$x` and `state.center$y` contain the centroid longitude and latitude points for each US state
- Centroid locations can be used to plot state labels

Maps

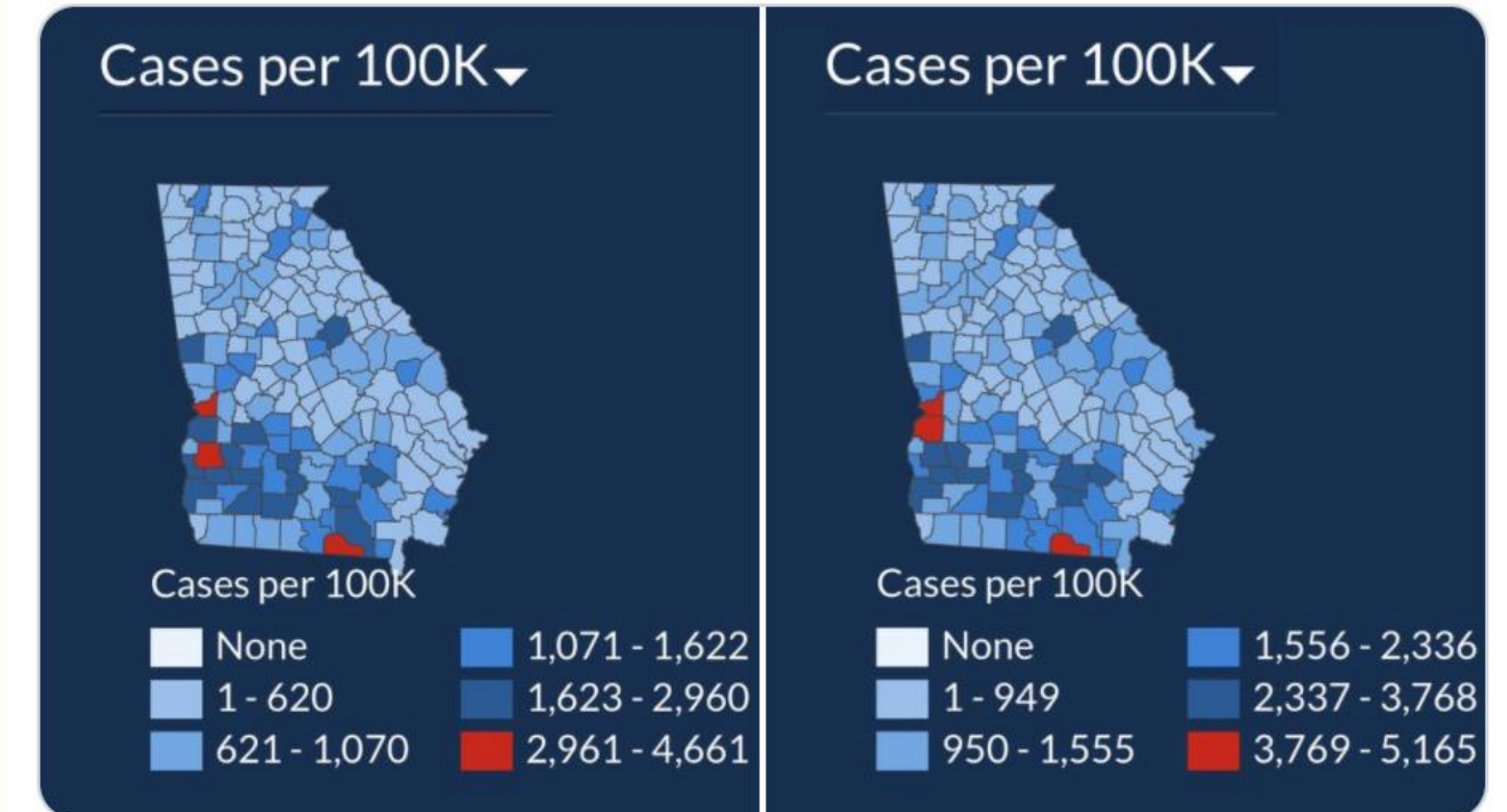
Case study: COVID-19

Heather Mattie, PhD

Misleading visualization

- In 2020 the Georgia Department of Public Health posted daily map visualizations of COVID-19 case counts in each county in Georgia
- On July 17, 2020, a tweet comparing the map on two different days pointed out an issue

In just 15 days the total number of [#COVID19](#) cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/

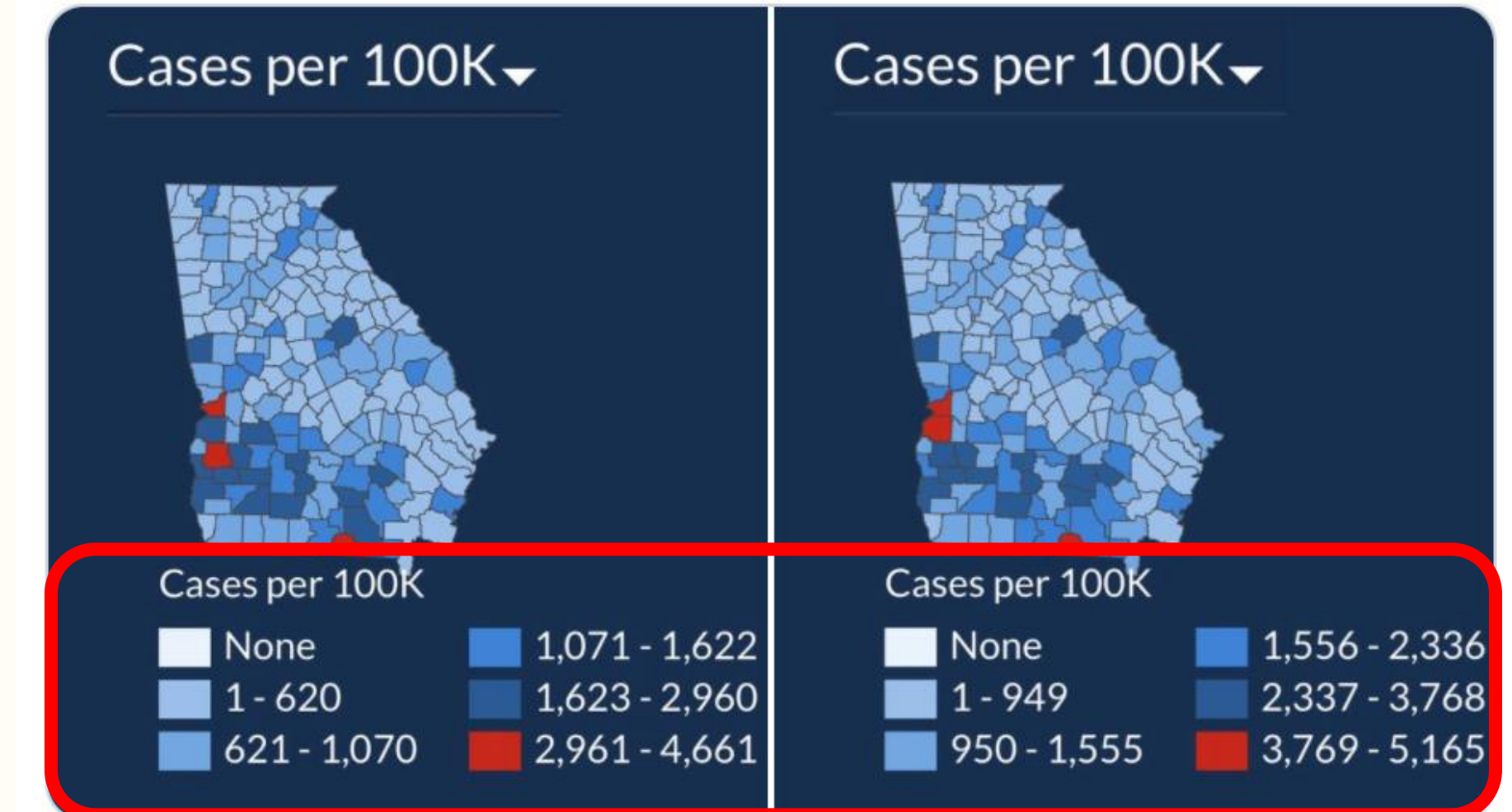


5:24 PM · Jul 17, 2020 · [Twitter for iPhone](#)

Misleading visualization

- In 2020 the Georgia Department of Public Health posted daily map visualizations of COVID-19 case counts in each county in Georgia
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In just 15 days the total number of [#COVID19](#) cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/



5:24 PM · Jul 17, 2020 · Twitter for iPhone

Data

We will need to combine three different datasets to recreate this plot

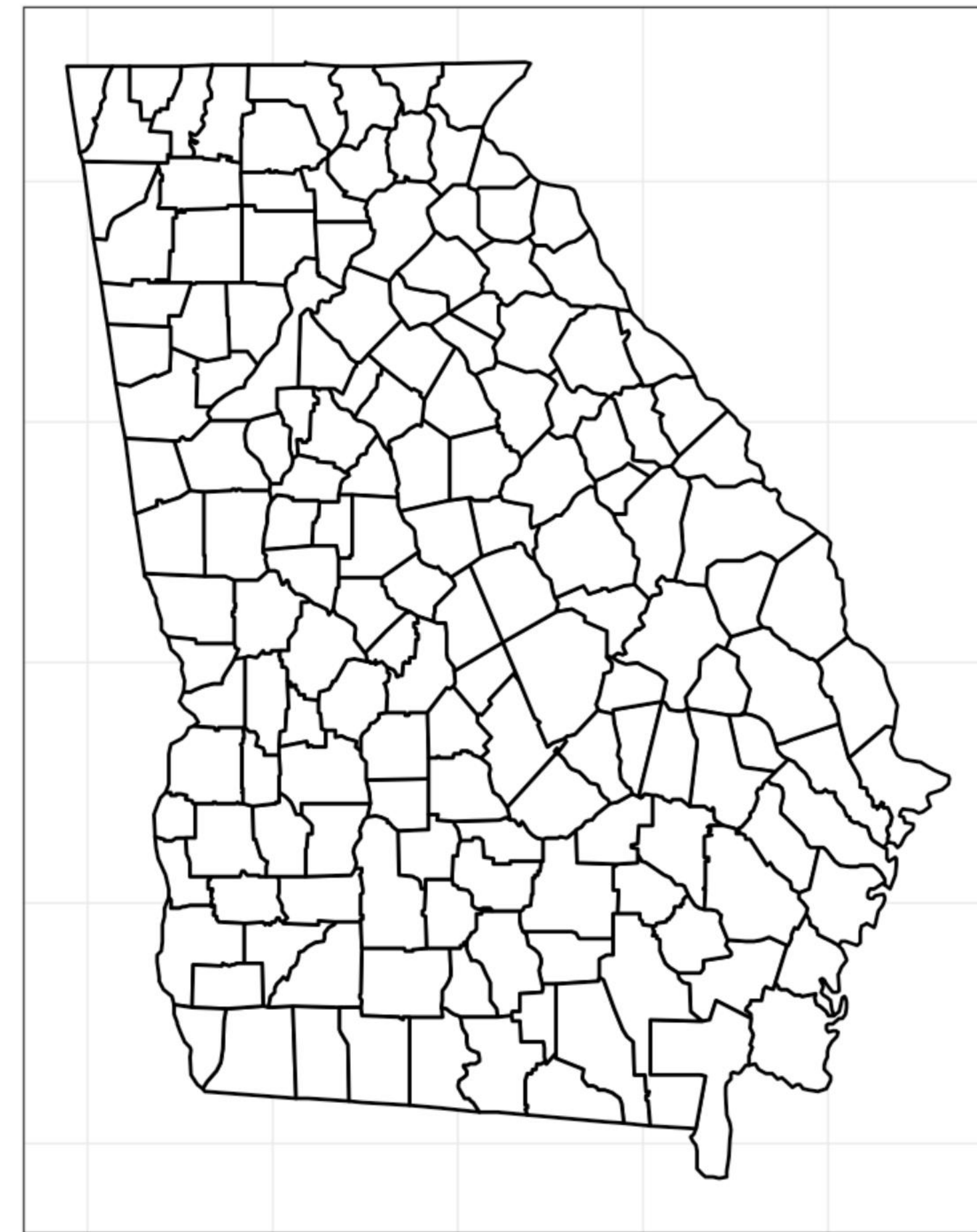
- Georgia county map data
 - From the **maps** package
- Daily COVID-19 case counts for each Georgia county
 - From New York Times COVID-19 data repository
- Population of each Georgia county
 - From the Georgia Department of Public Health (saved as csv)

Georgia county map data

```
GeorgiaCounty <- map_data("county", "georgia")  
head(GeorgiaCounty)
```

```
##           long      lat group order  region subregion  
## 1 -82.44862 31.94813     1     1 georgia  applying  
## 2 -82.42570 31.94813     1     2 georgia  applying  
## 3 -82.40852 31.94240     1     3 georgia  applying  
## 4 -82.39706 31.94240     1     4 georgia  applying  
## 5 -82.38560 31.93094     1     5 georgia  applying  
## 6 -82.35122 31.91948     1     6 georgia  applying
```

```
GeorgiaCounty %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(fill = "white", color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3)
```



COVID-19 data

nytimes / covid-19-data

Q Type / to search

+

<> Code

Issues 2

Pull requests

Projects

Security

Insights

covid-19-data

Public

Watch 317

Fork 3.4k

Star 7k

master

2 Branches

0 Tags

Go to file

Add file

<> Code

nyt-covid-19-bot

Updating data.

62ef34c · 2 years ago

4,101 Commits

.github/ISSUE_TEMPLATE	New data for 6/26.	5 years ago
colleges	update colleges readme to note it is not currently being u...	4 years ago
excess-deaths	Final update for excess deaths dataset.	5 years ago
live	Updating data.	2 years ago
mask-use	*NEW DATASET*: Estimates of mask-usage by county fro...	5 years ago
prisons	NEW: We're publishing files detailing the outbreak in priso...	4 years ago
rolling-averages	Updating data.	2 years ago
.gitignore	excess deaths	5 years ago
LICENSE	Update license and citation year	5 years ago

About

A repository of data on coronavirus cases and deaths in the U.S.

[www.nytimes.com/interactive/2020/u...](#)

covid-19

Readme

View license

Activity

Custom properties

7k stars

317 watching

3.4k forks

Report repository

Releases



COVID-19 data

```
url <- "https://raw.githubusercontent.com/nytimes/covid-19-data/master/us-counties.csv"
cases <- read_csv(url)
head(cases)
```

```
## # A tibble: 6 × 6
##   date      county    state    fips  cases deaths
##   <date>    <chr>    <chr>    <chr> <dbl>  <dbl>
## 1 2020-01-21 Snohomish Washington 53061     1      0
## 2 2020-01-22 Snohomish Washington 53061     1      0
## 3 2020-01-23 Snohomish Washington 53061     1      0
## 4 2020-01-24 Cook      Illinois   17031     1      0
## 5 2020-01-24 Snohomish Washington 53061     1      0
## 6 2020-01-25 Orange     California 06059     1      0
```

COVID-19 data

```
dates <- c(ymd("2020-07-02", "2020-07-17"))
georgia_cases <- cases %>% filter(state == "Georgia", date %in% dates)
head(georgia_cases)
```

```
## # A tibble: 6 × 6
##   date      county  state  fips  cases  deaths
##   <date>    <chr>   <chr> <chr> <dbl>  <dbl>
## 1 2020-07-02 Appling  Georgia 13001   266    14
## 2 2020-07-02 Atkinson Georgia 13003   155     2
## 3 2020-07-02 Bacon    Georgia 13005   254     4
## 4 2020-07-02 Baker    Georgia 13007    43     3
## 5 2020-07-02 Baldwin Georgia 13009   543    34
## 6 2020-07-02 Banks    Georgia 13011   140     1
```

Georgia county population data

```
pop_data <- read_csv("county_pop.csv")
head(pop_data)
```

```
## # A tibble: 6 × 12
##   county_name cases county_id `State FIPS code` `County FIPS code` population
##   <chr>      <dbl> <chr>      <chr>          <chr>          <dbl>
## 1 Appling      1054 US-13001  13              1              18561
## 2 Atkinson      447 US-13003  13              3              8330
## 3 Bacon         602 US-13005  13              5             11404
## 4 Baker          84 US-13007  13              7              3116
## 5 Baldwin     2132 US-13009  13              9             44428
## 6 Banks         501 US-13011  13             11             19982
## # i 6 more variables: hospitalization <dbl>, deaths <dbl>, `case rate` <dbl>,
## #   `death rate` <dbl>, `14 day case rate` <dbl>, `14 day cases` <dbl>
```

Joining data frames

1. Join the county population data to the COVID-19 cases data

```
pop_data <- pop_data %>% select(county_name, population)
```

```
data_full <- left_join(georgia_cases, pop_data, by = c("county" = "county_name"))  
head(data_full)
```

```
## # A tibble: 6 × 7  
##   date      county  state  fips  cases  deaths  population  
##   <date>    <chr>   <chr> <chr> <dbl>  <dbl>    <dbl>  
## 1 2020-07-02 Appling  Georgia 13001   266     14    18561  
## 2 2020-07-02 Atkinson Georgia 13003   155      2     8330  
## 3 2020-07-02 Bacon    Georgia 13005   254      4    11404  
## 4 2020-07-02 Baker    Georgia 13007    43      3     3116  
## 5 2020-07-02 Baldwin  Georgia 13009   543     34    44428  
## 6 2020-07-02 Banks    Georgia 13011   140      1    19982
```

Joining data frames

- 1. Join the county population data to the COVID-19 cases data
- 2. Calculate cases per 100,000 population

```
data_full <- data_full %>% mutate(rate = 100000*(cases/population))
head(data_full)
```

```
## # A tibble: 6 × 8
```

	date	county	state	fips	cases	deaths	population	rate
	<date>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	2020-07-02	Appling	Georgia	13001	266	14	18561	1433.
## 2	2020-07-02	Atkinson	Georgia	13003	155	2	8330	1861.
## 3	2020-07-02	Bacon	Georgia	13005	254	4	11404	2227.
## 4	2020-07-02	Baker	Georgia	13007	43	3	3116	1380.
## 5	2020-07-02	Baldwin	Georgia	13009	543	34	44428	1222.
## 6	2020-07-02	Banks	Georgia	13011	140	1	19982	701.



Joining data frames

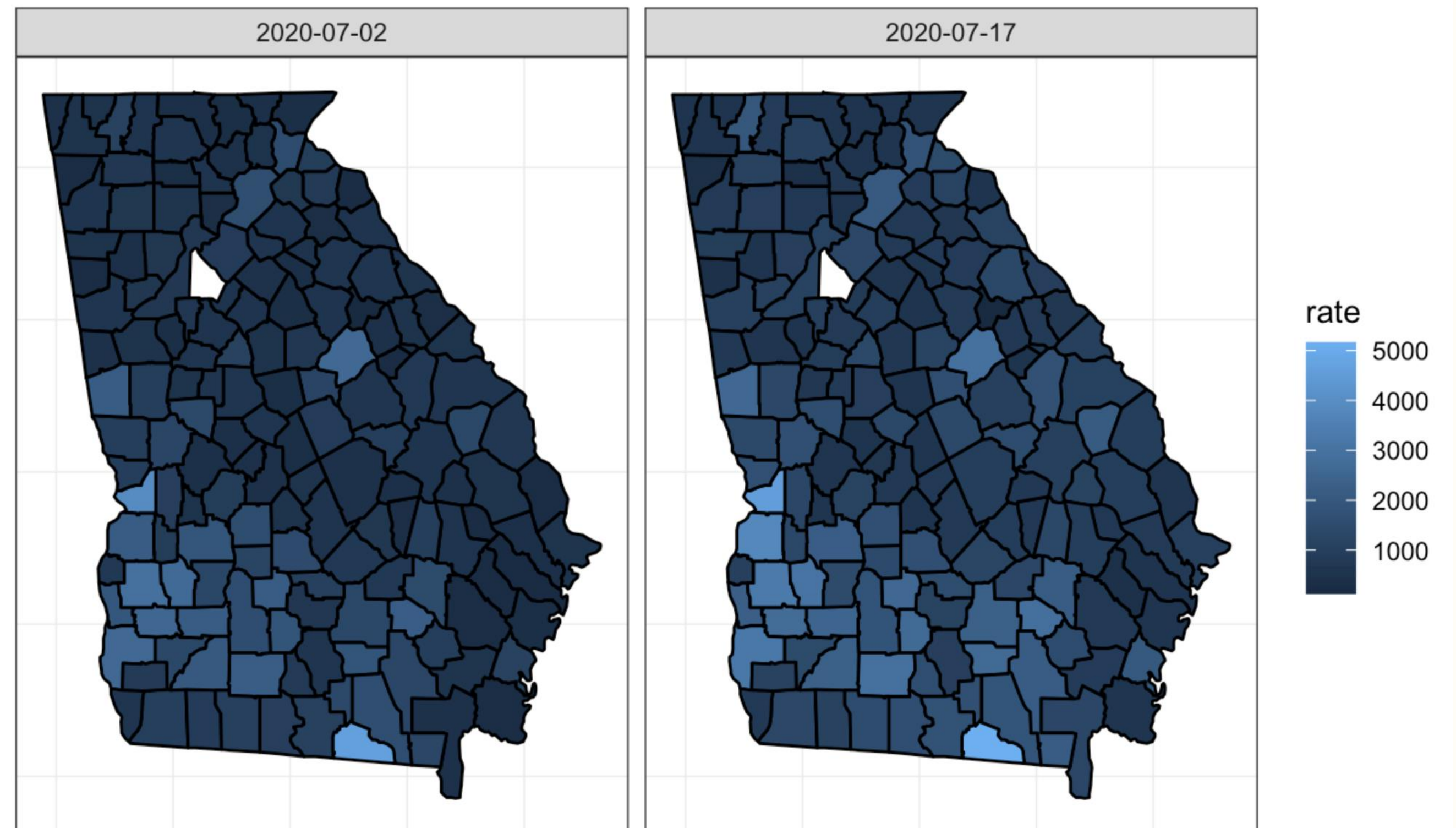
- 1. Join the county population data to the COVID-19 cases data
- 2. Calculate cases per 100,000 population
- 3. Join to map data

```
georgia_map <- data_full %>% mutate(county = str_to_lower(county)) %>%  
  left_join(GeorgiaCounty, by = c("county" = "subregion"))  
  
head(georgia_map)
```

```
## # A tibble: 6 × 13  
##   date      county state fips  cases deaths population  rate  long  lat group  
##   <date>    <chr>  <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>  
## 1 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 2 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 3 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 4 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 5 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 6 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## # i 2 more variables: order <int>, region <chr>
```

Data inconsistencies

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```

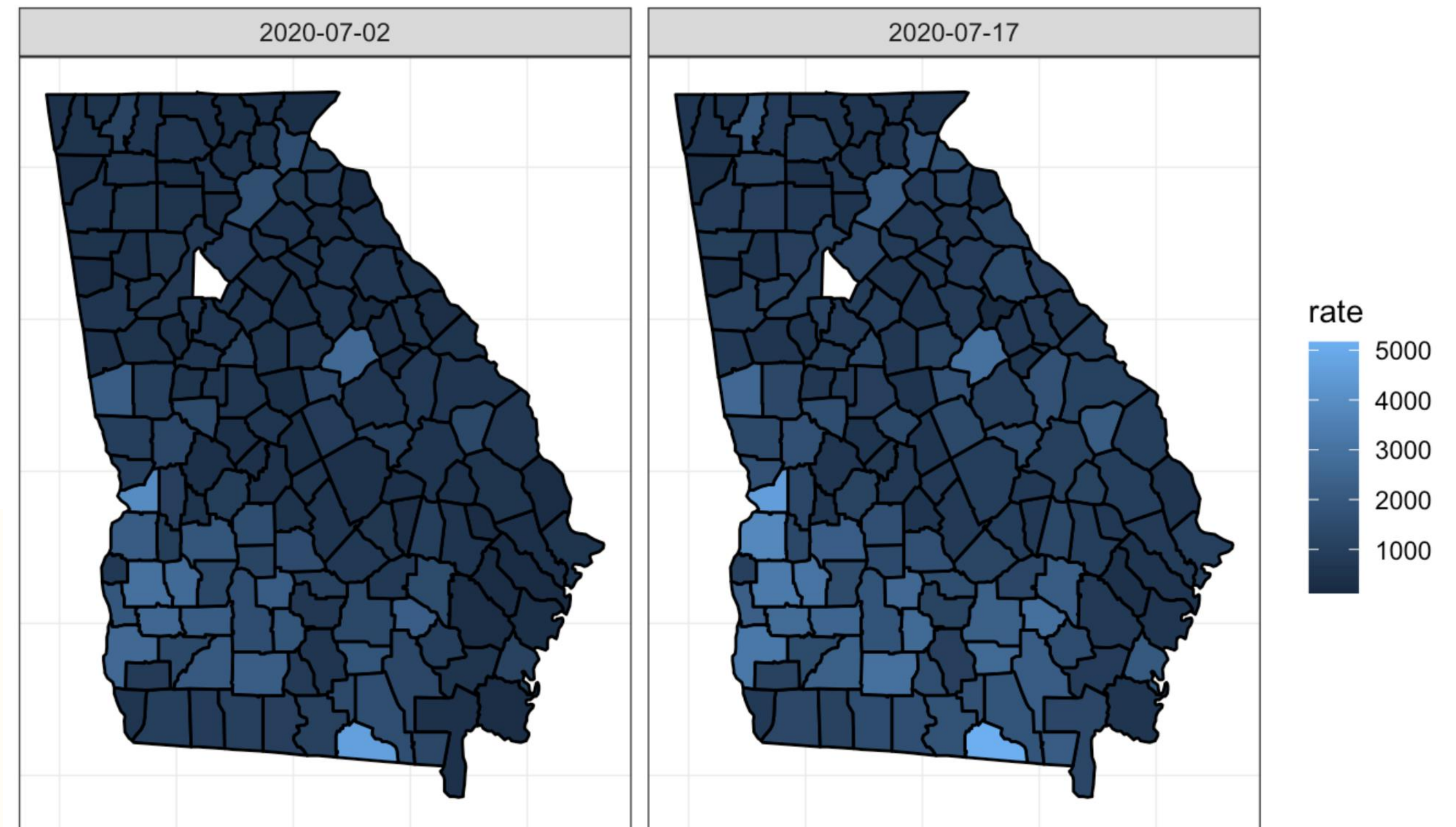


Data inconsistencies

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```

Problem: De Kalb county

Saved as “**de kalb**” in
the **GeorgiaCounty** data frame, but as
“**DeKalb**” in the in the **georgia_cases**
data frame



Data inconsistencies

First solution attempt:

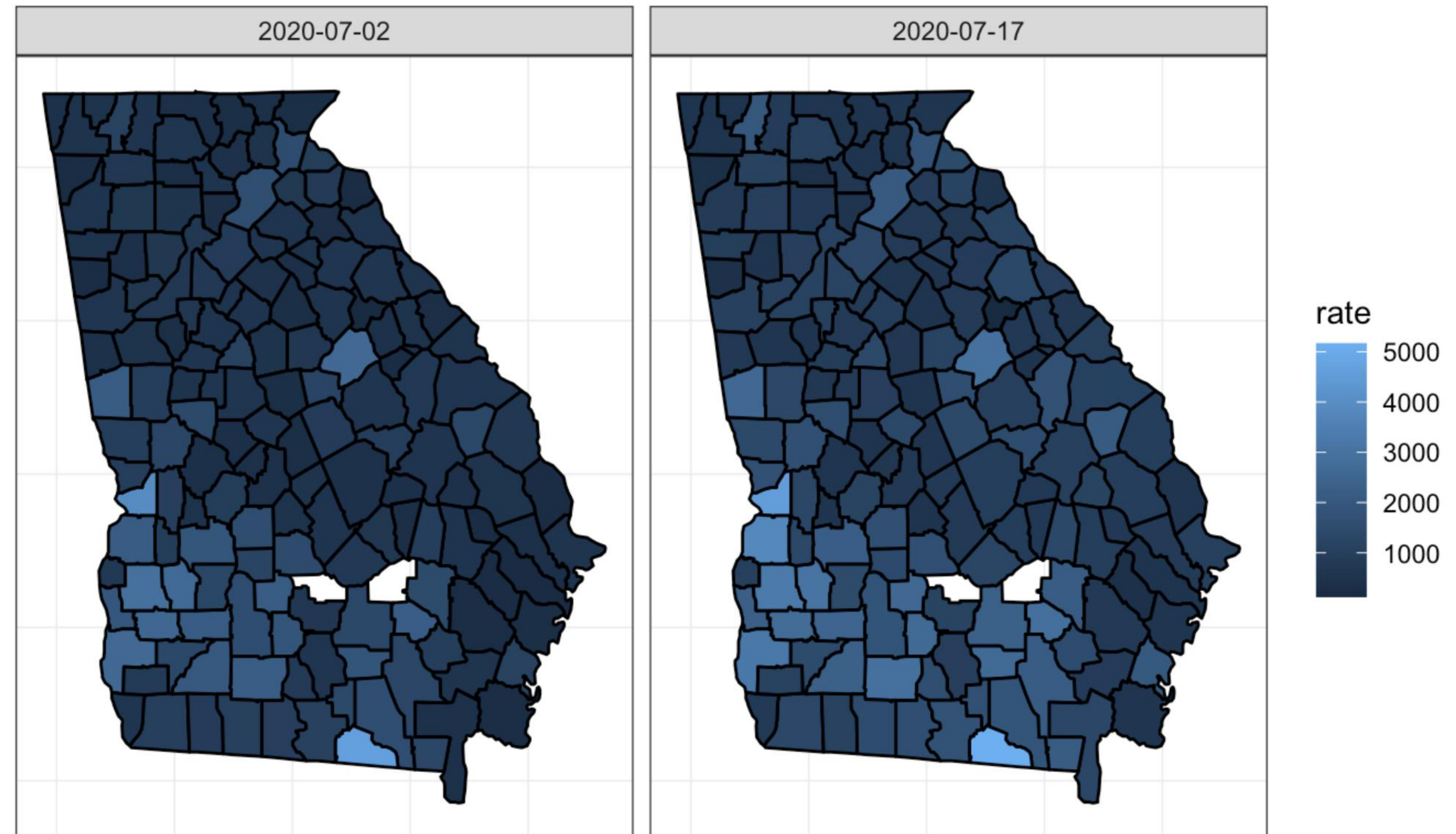
1. Remove all spaces in county names using the **str_replace** function
2. Convert all county names to lowercase using the **str_to_lower** function

```
GeorgiaCounty <- GeorgiaCounty %>%  
  mutate(subregion = str_replace(subregion, " ", ""))  
  
georgia_map <- data_full %>%  
  mutate(county = str_to_lower(county)) %>%  
  left_join(GeorgiaCounty, by = c("county" = "subregion"))  
  
head(georgia_map)
```

```
## # A tibble: 6 × 13  
##   date      county state fips  cases deaths population  rate  long  lat group  
##   <date>    <chr>  <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>  
## 1 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 2 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 3 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 4 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 5 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 6 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## # i 2 more variables: order <int>, region <chr>
```

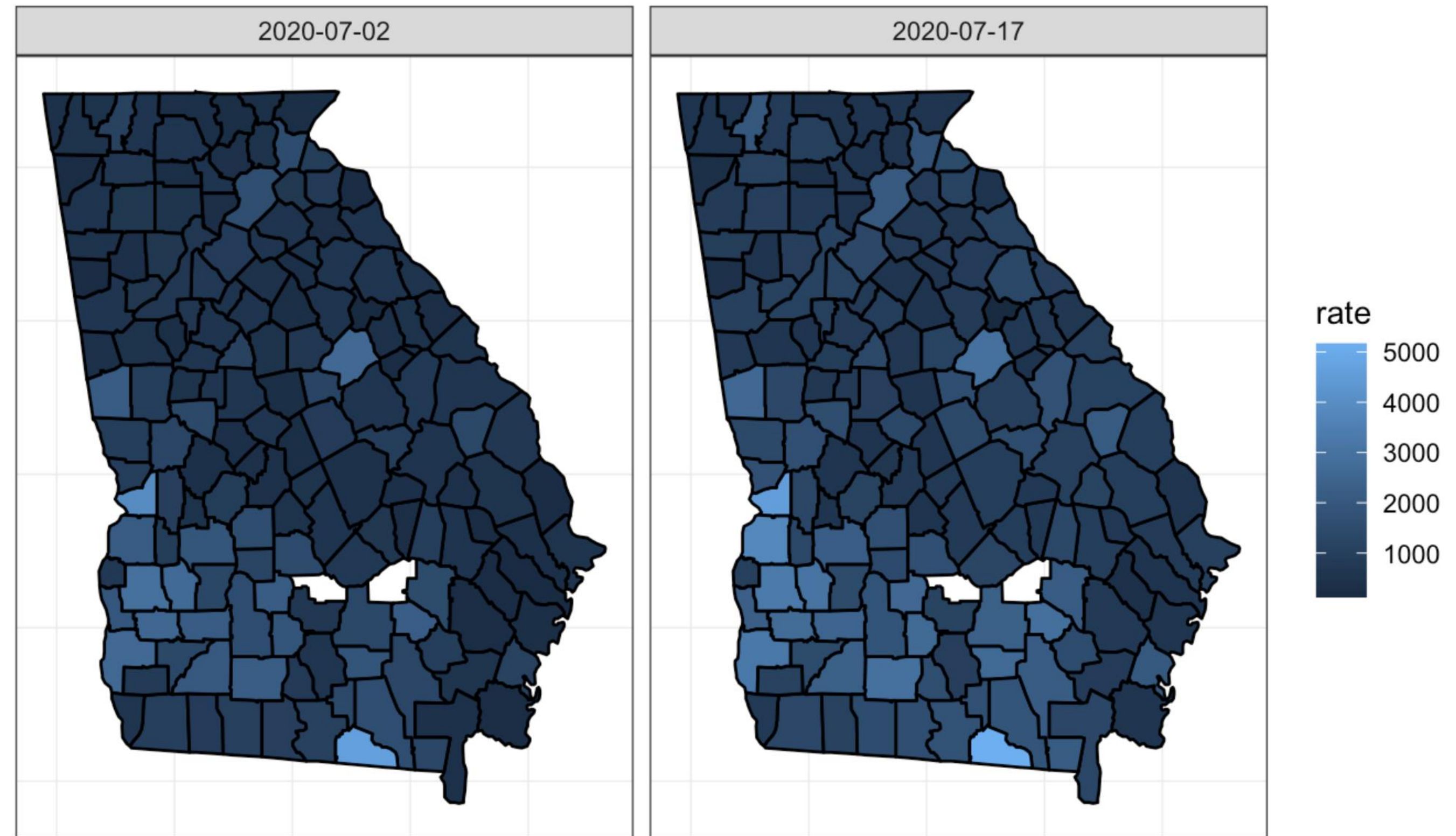
Data inconsistencies

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```



Data inconsistencies

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```



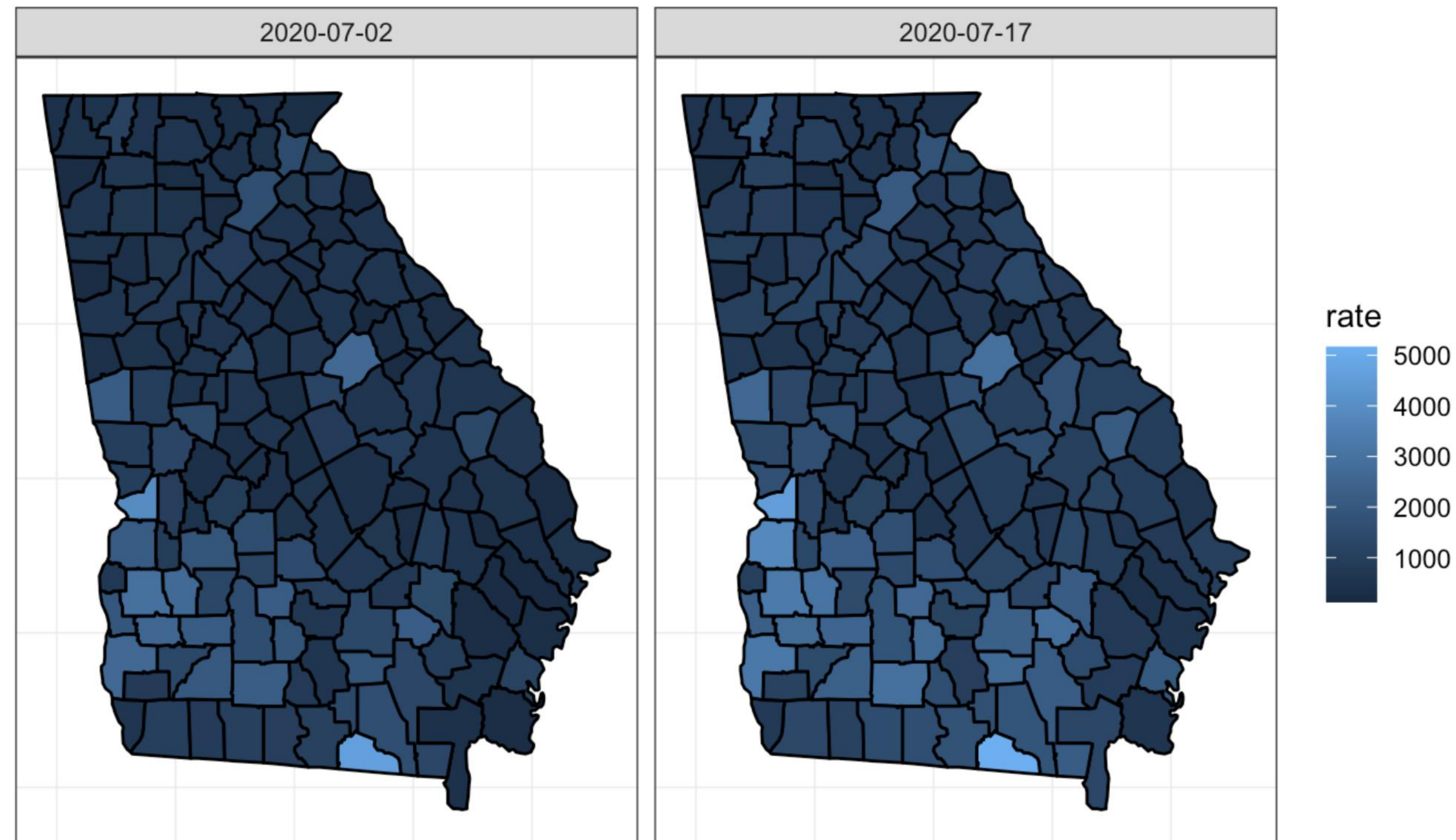
Problem: **Ben Hill** and **Jeff Davis** counties

Both data frames had a space between the two names but the space was removed for **GeorgiaCounty** and not for **georgia_cases**

Data inconsistencies

```
data_full <- data_full %>%  
  mutate(county = str_replace(county, " ", ""))  
  
georgia_map <- data_full %>%  
  mutate(county = str_to_lower(county)) %>%  
  left_join(GeorgiaCounty, by = c("county" = "subregion"))
```

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```



Customizing legends

```
georgia_map <- georgia_map %>%  
  mutate(manual_fill = cut(rate, breaks = c(0, 620, 1070, 1622, 2960, Inf),  
    labels = c("1-620", "621-1,070", "1,071-1,622",  
      "1,623-2,960", ">2,960"),  
    right = TRUE))  
  
head(georgia_map)
```

```
## # A tibble: 6 × 14  
##   date      county state fips  cases deaths population  rate  long  lat group  
##   <date>    <chr>  <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>  
## 1 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 2 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 3 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 4 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 5 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 6 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## # i 3 more variables: order <int>, region <chr>, manual_fill <fct>
```

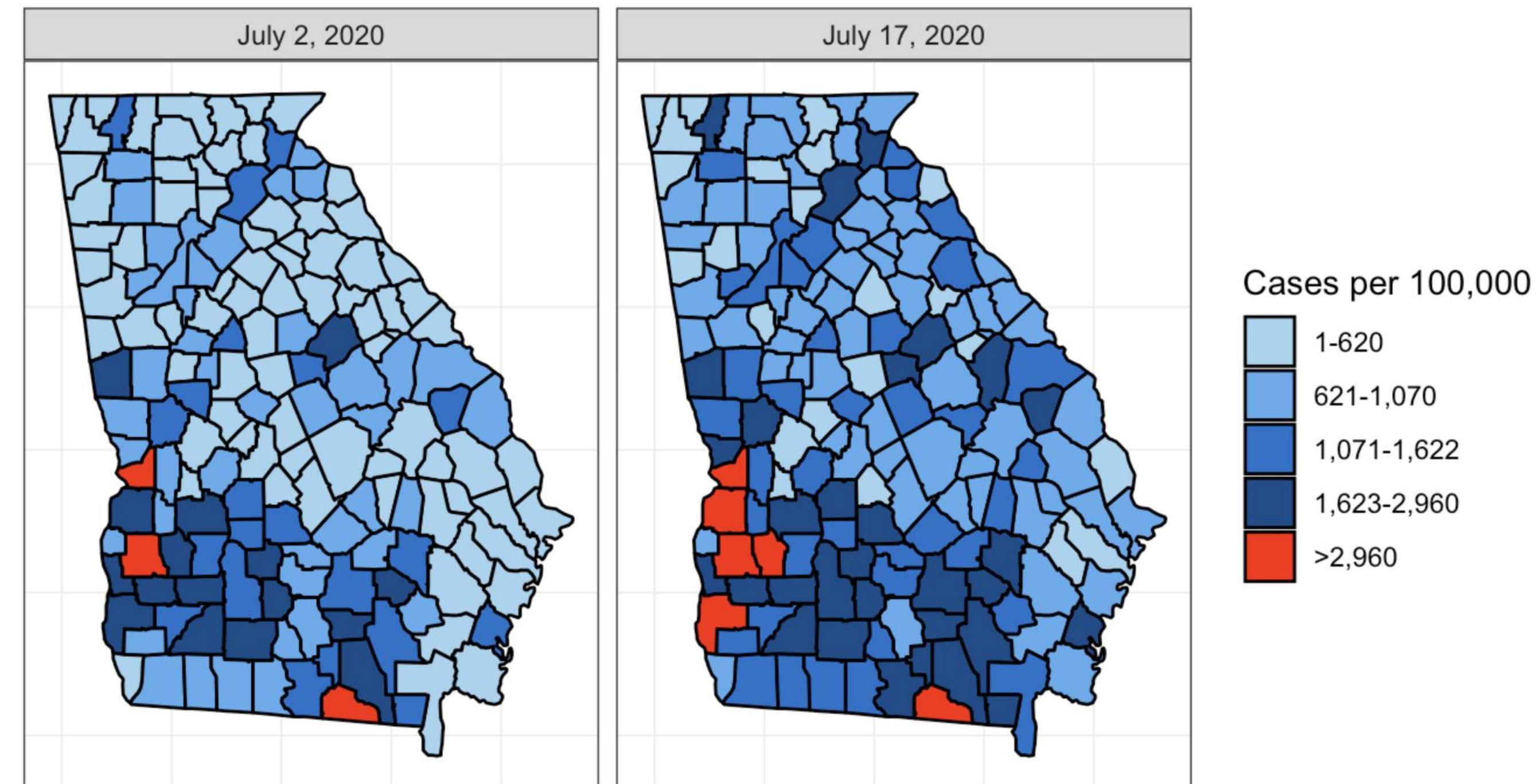
Customizing color palettes

```
pal <- c("lightskyblue2", "steelblue2", "dodgerblue3", "dodgerblue4", "red")
```

```
date.labs <- c("July 2, 2020", "July 17, 2020")  
names(date.labs) <- c("2020-07-02", "2020-07-17")
```

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(aes(fill = manual_fill), color = "black") +  
  scale_fill_manual(name = "Cases per 100,000", values = pal) +  
  coord_fixed(1.3) +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  ggtitle("COVID-19 Cases per 100K") +  
  facet_grid(. ~ date, labeller = labeller(date = date.labs))
```

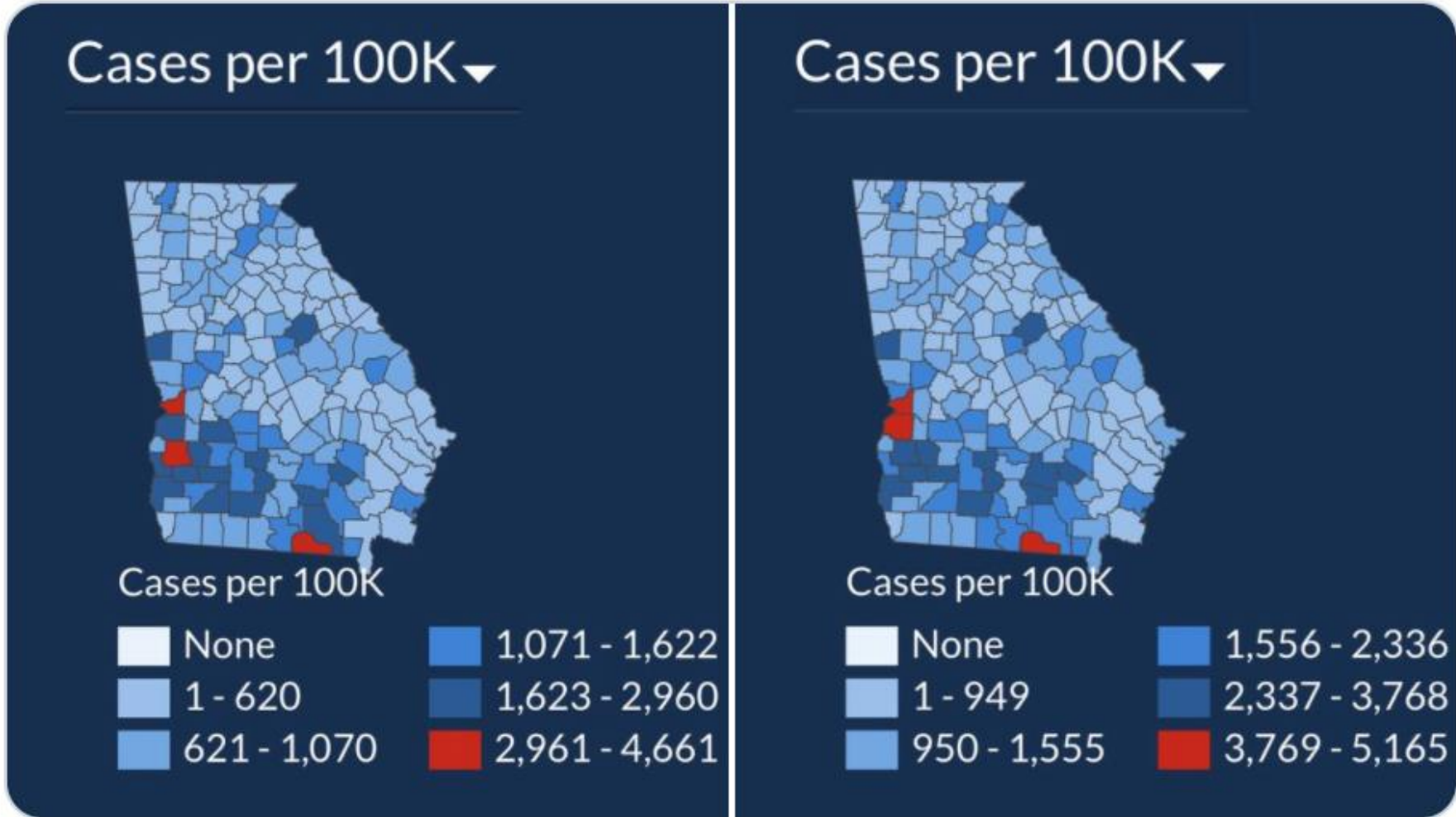
COVID-19 Cases per 100K



Plot comparison

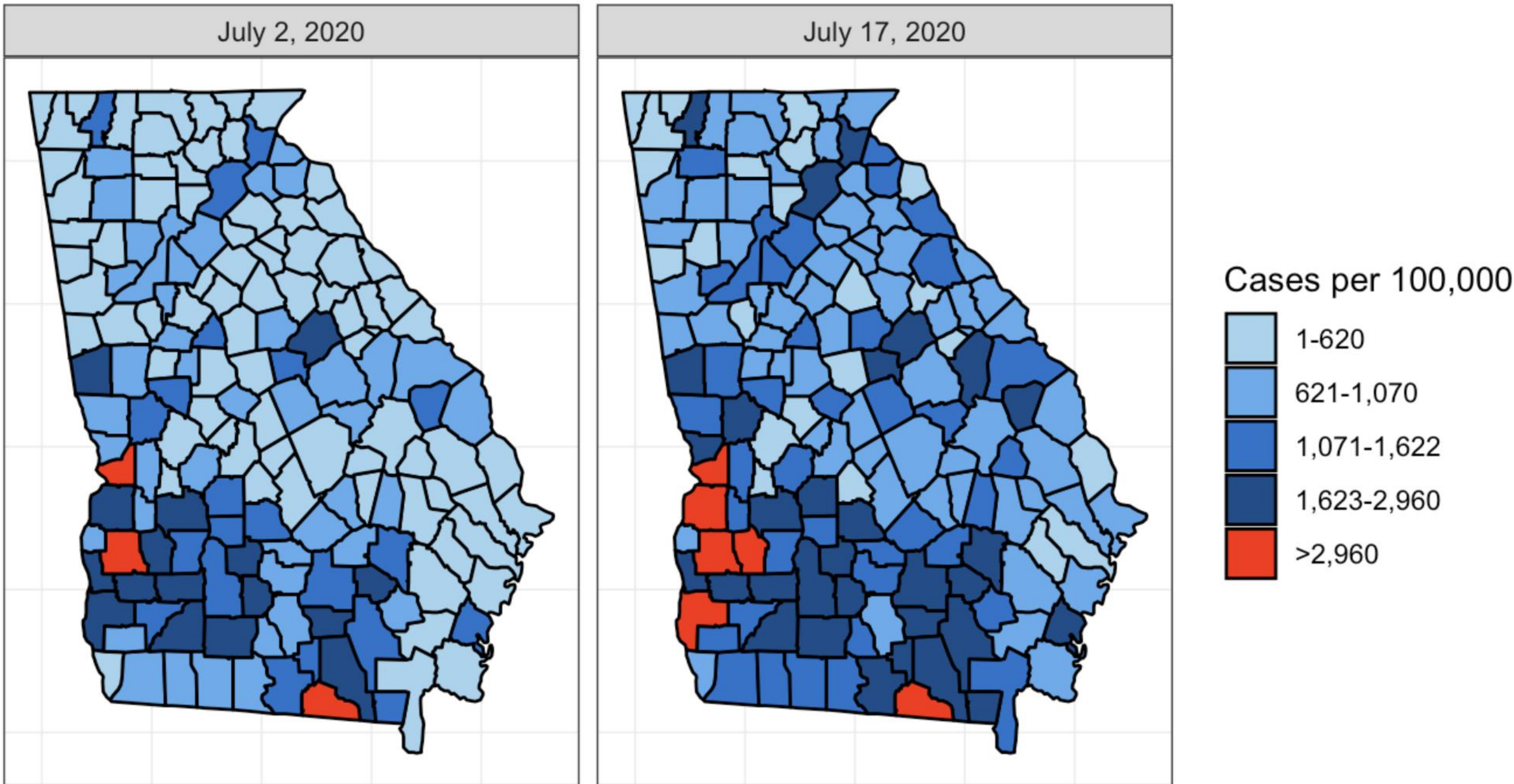


In just 15 days the total number of [#COVID19](#) cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/



5:24 PM · Jul 17, 2020 · [Twitter for iPhone](#)

COVID-19 Cases per 100K



Summary

- Plots of maps can be misleading, especially when plotting changes over time
- There should only be one legend per plot if subplots are related
- Color palettes can be customized